Overview

A. How machine learning is used in industry

Machine learning has become one of the most important tools used in the tech industry. It powers everything from self-driving cars to virtual assistants. Nearly every company either has machine learning in their tech stack or is planning to incorporate machine learning in the near future.

Many people who have taken a college course or an online course in machine learning are surprised at how different the machine learning process is in industry. Very seldom in industry will people need to know much of the theory or mathematics behind machine learning; rather, the main focus is in applying existing algorithms and models to new problems and datasets.

Furthermore, the majority of work done by machine learning engineers in industry has nothing to do with actually building out machine learning models. Rather, most of the time spent on machine learning projects is in analyzing datasets, creating a data pipeline, and interpreting model results.

B. What this course will provide

After taking this course, you'll be able to take on the same types of projects done by industry machine learning engineers. Specifically, you will be able to:

- Process and analyze datasets.
- Create an efficient data pipeline for feeding input data into machine learning models.
- Develop robust machine learning models for both classification and regression tasks.
- Interpret model results with respect to the project domain.

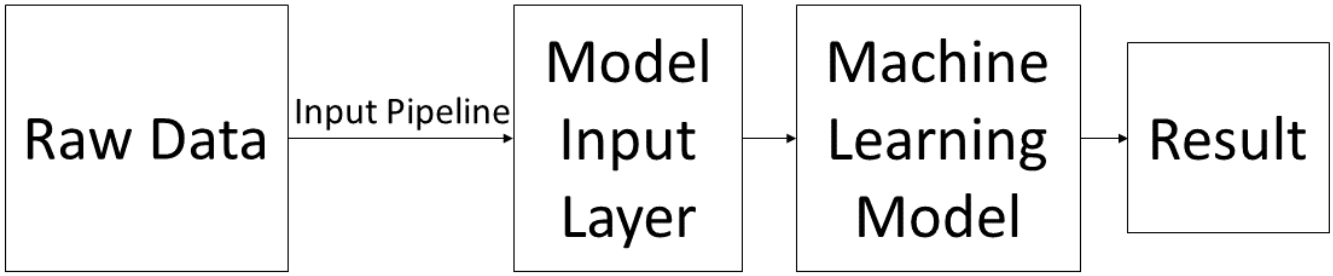
Introduction

In this section of the course you will be using TensorFlow to load and store data. Specifically, you will use protocol buffers to efficiently store large amounts of data and then use TensorFlow's `tf.data` API to load the data. We will be using a dataset comprised of college student data to test your code.

A. Input pipeline

Machine learning models are almost always trained on extremely large datasets. Depending on the application, a training dataset can have hundreds of thousands, or even millions, of training examples. The sheer size of these datasets makes it incredibly important to store the data in an efficient manner.

The process in which data is loaded from files and fed into a machine learning model is known as the *input pipeline*. Since the input pipeline handles a large amount of data for machine learning projects, we need it to be as efficient as possible.



Input pipeline for a machine learning model.

A flexible and efficient format for storing large amounts of data is Google's protocol buffer. The protocol buffer is similar to JSON and XML (another feature-based data format), but uses less space and is faster to process. When used with TensorFlow, protocol buffers make the input pipeline for large datasets much more streamlined.

B. Loading data

One of the main reasons why protocol buffers work so well with TensorFlow is the ease with which they can be loaded as input into a machine learning model. Since TensorFlow is also a Google-developed product, the TensorFlow API contains functions that make it simple to quickly load data from a protocol buffer. This is why most of the official TensorFlow open-source models use protocol buffers for storing and loading data.

In particular, the `tf.data` API provides us with all the tools necessary to create an efficient input pipeline. While it works very well with protocol buffers (which are used to store feature-based data), it can also be used effectively with NumPy data. In the following chapters, you'll create input pipelines from both protocol buffers and NumPy data.

Protocol Buffer

Discover the benefits of using protocol buffers in TensorFlow.

Chapter Goals:

- Learn how protocol buffers are used in TensorFlow
- Implement a function to convert a Python dictionary to a `tf.train.Example` object

A. TensorFlow protocol buffer

Since protocol buffers use a structured format when storing data, they can be represented with Python classes. In TensorFlow, the `tf.train.Example` class represents the protocol buffer used to store data for the input pipeline.

Each individual `tf.train.Example` object describes data for a single dataset observation (e.g. a single row in a data table). We convert raw data to a protocol buffer by initializing a `tf.train.Example` object with the data's values.

When we initialize a `tf.train.Example` object, we need to set that object's `features` argument to a `tf.train.Features` object. The `tf.train.Features` class is initialized by setting the `feature` field to a dictionary that maps feature names to feature values.

```
import tensorflow as tf

features = tf.train.Features(feature=f_dict)  # f_dict is a dict
ex = tf.train.Example(features=features)
print(repr(ex))
```

The code above initializes a `tf.train.Example` object. Note that `f_dict` is a dictionary mapping feature names to feature values. We discuss how to create the feature dictionary later in this chapter.

the feature dictionary later in this chapter.

B. Features

Each feature value is represented by a `tf.train.Feature` object, which is initialized with exactly one of the following fields:

- `int64_list`, for integer data values. Set with a `tf.train.Int64List` object.
- `float_list`, for floating point data values. Set with a `tf.train.FloatList` object.
- `bytes_list`, for byte string data values. Set with a `tf.train.BytesList` object.

The code below creates `tf.train.Feature` objects from data values. The `encode` function used in the last example converts the string to bytes, so the type is compatible with `tf.train.BytesList`.

```
import tensorflow as tf

int_f = tf.train.Feature(
    int64_list=tf.train.Int64List(value=[1, 2]))
print(repr(int_f) + '\n')

float_f = tf.train.Feature(
    float_list=tf.train.FloatList(value=[-8.2, 5]))
print(repr(float_f) + '\n')

bytes_f = tf.train.Feature(
    bytes_list=tf.train.BytesList(value=[b'\xff\xcc', b'\xac']))
print(repr(bytes_f) + '\n')

str_f = tf.train.Feature(
    bytes_list=tf.train.BytesList(value=['joe'.encode()]))
print(repr(str_f) + '\n')
```

Note that the `value` field for the `tf.train.Int64List`, `tf.train.FloatList`, and `tf.train.BytesList` classes must be an iterable (e.g. list, NumPy array, or pandas Series). If a feature only has one data value, we would pass in an iterable containing the single value.

With the `tf.train.Feature` objects, we can create the dictionary that's used to initialize a `tf.train.Features` object.

```
import tensorflow as tf
```

```
f_dict = {
    'int_vals': int_f,

    'float_vals': float_f,
    'bytes_vals': bytes_f,
    'str_vals': str_f
}

features = tf.train.Features(feature=f_dict)

print(repr(features))
```

C. Bytes and text

When dealing with datasets containing bytes (e.g. images or videos) or text (e.g. articles or sentences), it is beneficial to first read all the data files and then store the read data in the `bytes_list` field of a `tf.train.Feature`. This saves us from having to open each individual file within our input pipeline, which can drastically improve efficiency.

task.py

story.txt

```
import tensorflow as tf

with open('story.txt') as f:
    words = f.read().split()

encw = [w.encode() for w in words]
words_feature = tf.train.Feature(
    bytes_list=tf.train.BytesList(value=encw))
print(repr(words_feature))

with open('img.jpg', 'rb') as f:
    img_bytes = f.read()

img_feature = tf.train.Feature(
    bytes_list=tf.train.BytesList(value=[img_bytes]))
print(repr(img_feature))
```

Time to Code!

In this chapter, you'll be completing the `dict_to_example` function, which converts a regular dictionary into a `tf.train.Example` object.

```
import tensorflow as tf
```

```
def dict_to_example(data_dict, config):
    feature_dict = {}
    for feature_name, value in data_dict.items():

        feature_config = config[feature_name]
        shape = feature_config['shape']
        if shape == () or shape == []:
            value = [value]
        value_type = feature_config['type']
        if value_type == 'int':
            feature_dict[feature_name] = make_int_feature(value)
        elif value_type == 'float':
            feature_dict[feature_name] = make_float_feature(value)
        elif value_type == 'string' or value_type == 'bytes':
            feature_dict[feature_name] = make_bytes_feature(
              value, value_type)
    features = tf.train.Features(feature=feature_dict)
    return tf.train.Example(features=features)
```

The `dict_to_example` function uses 3 helper functions: `make_int_feature`, `make_float_feature`, and `make_bytes_feature`. You will be creating each of these helper functions.

The `make_int_feature` function returns a `tf.train.Feature` initialized with the `int64_list` field.

Return a `tf.train.Feature` object with the `int64_list` attribute set equal to a `tf.train.Int64List` object (initialized with `value`).

```
import tensorflow as tf

def make_int_feature(value):
    # CODE HERE
    pass
```

The `make_float_feature` function returns a `tf.train.Feature` initialized with the `float_list` field.

Return a `tf.train.Feature` object with the `float_list` attribute set equal to a `tf.train.FloatList` object (initialized with `value`).

```
import tensorflow as tf

def make_float_feature(value):
    # CODE HERE
    pass
```

The `make_float_feature` function returns a `tf.train.Feature` initialized with the `bytes_list` field. Note that if the `value_type` is `'string'` we need to encode each of the elements in `value`.

If `value_type` is equal to `'string'`, encode each element in `value` (and keep the resultant list as `value`).

Return a `tf.train.Feature` object with the `bytes_list` attribute set equal to a `tf.train.BytesList` object (initialized with `value`).

```
import tensorflow as tf

def make_bytes_feature(value, value_type):
    # CODE HERE
    pass
```

TFRecords

Learn how protocol buffers are stored in TFRecords files.

Chapter Goals:

- Learn how to write serialized protocol buffers to TFRecords files
- Implement a function that writes a list of feature data to a TFRecords file

A. Serialization

After creating a `tf.train.Example` protocol buffer, we normally store it in a file. To do this, we first have to *serialize* the object, i.e. convert it to a byte string which can be written to a file. The way we serialize a `tf.train.Example` object is through its `SerializeToString` method.

```
import tensorflow as tf

ex = tf.train.Example(features=tf.train.Features(feature=f_dict))

print(repr(ex))

ser_ex = ex.SerializeToString()
print(ser_ex)
```

B. Writing to data files

We store serialized `tf.train.Example` protocol buffers in special files called TFRecords files. The simple way to write to a TFRecords file is through a `TFRecordWriter`.

```
import tensorflow as tf

writer = tf.python_io.TFRecordWriter('out.tfrecords')
writer.write(ser_ex)
writer.close()
```

The `TFRecordWriter` is initialized with the output file that it writes to. In our example, we wrote to `'out.tfrecords'`.

The `write` function takes in a byte string and writes that byte string to the end of the output file. After we're done writing to the output file, we close the file using the `close` function.

You can also write multiple serialized `tf.train.Example` objects to a single file, as long as the file is open.

```
import tensorflow as tf

# Writing 3 Example objects to the same file
writer = tf.python_io.TFRecordWriter('out.tfrecords')
writer.write(ser_ex1)
writer.write(ser_ex2)
writer.write(ser_ex3)
writer.close()
```

Features

Understand how data features are represented in TensorFlow.

Chapter Goals:

- Learn what an Example spec is and what it is used for
- Understand the two different feature configuration classes in TensorFlow
- Implement a function that creates an Example spec from a feature configuration

A. Example spec

When we read from a TFRecords file for our input pipeline, we get back the serialized versions of the protocol buffers. What we really want, though, are easily accessible data values for each feature from the original protocol buffer. In order to achieve this, we need an *Example spec*.

An Example spec is a dictionary that maps the feature names from the original `tf.train.Example` protocol buffer to particular configuration classes for each feature. Specifically, a feature is in exactly one of two configuration classes: `VarLenFeature` or `FixedLenFeature`.

B. `VarLenFeature` class

A `VarLenFeature` specifies a feature whose data value list has variable length, meaning each protocol buffer can have a different number of values for the feature.

```
import tensorflow as tf

# Two people represented by the same protocol buffer
# "jobs" is a VarLenFeature
print(repr(person1))  # 1 job
print(repr(person2))  # 2 jobs
```

In the above code example, the `"jobs"` feature is a `VarLenFeature`. For `person1`, the data value list has length 1, but for person2 the data value list has length 2.

Other common features that fall into the `VarLenFeature` class are text features (e.g. articles with different word counts), features that depend on longevity (e.g. salary history for employees at a company), and features that allow optional or incomplete values.

C. `FixedLenFeature` class

In contrast with `VarLenFeature`, a `FixedLenFeature` has a fixed length for its data value list. This means each protocol buffer should have exactly the same number of values for a feature in this class.

```
import tensorflow as tf

# Two people represented by the same protocol buffer
# "name" and "salary" are both FixedLenFeature
print(repr(person1))
print(repr(person2))
```

In this example, both the `"name"` and `"salary"` features are part of the `FixedLenFeature` configuration class. The `"name"` feature has value length 1 while the `"salary"` feature has value length 2 (hourly and monthly salaries).

Other common features that fall into the `FixedLenFeature` class are statistics over a fixed time range, features that take single values (e.g. name, birthplace, etc.), and rigidly structured features.

D. Class configuration

When we use either a `VarLenFeature` or `FixedLenFeature`, we always have to set the TensorFlow datatype. For the Example spec, this datatype will either be `tf.int64`, `tf.float32`, or `tf.string`.

In addition to the datatype, the `FixedLenFeature` class also takes in the fixed shape and an optional default value. The default value is used in place of any

missing values in the feature (e.g. if the shape specifies 4 values but the protocol buffer only has 2).

```
import tensorflow as tf

name = tf.FixedLenFeature((), tf.string)
jobs = tf.VarLenFeature(tf.string)
salary = tf.FixedLenFeature(2, tf.int64, default_value=0)
example_spec = {
    'name': name,
    'jobs': jobs,
    'salary': salary
}

print(example_spec)
```

In the code example above, we created an Example spec that contains two `FixedLenFeature` features and one `VarLenFeature` feature. The `VarLenFeature` class takes in the datatype as its only argument. The `FixedLenFeature` class takes in the shape as the first argument, datatype as the second argument, and optional default value as the third argument.

Note that the `()` represents a single value shape. This actually differs from using `1` as the shape argument, for reasons we'll discuss in upcoming chapters.

Time to Code!

In this chapter you'll be completing the `create_example_spec` function. The function creates an Example spec from a configuration dictionary `config`, which maps feature names to their configurations (similar to the one used in `dict_to_example` from chapter 2).

You'll specifically be creating a helper function called `make_feature_config` which returns either a `VarLenFeature` or `FixedLenFeature` depending on the `shape` of the feature.

Create an `if...else` block where the `if` condition checks that `shape` is `None`.

If the feature's shape is `None`, that means we don't know the shape. In other words, the feature can be of variable length. In this case, we use `VarLenFeature` in our Example spec.

In the `if` block, set `feature` equal to `tf.VarLenFeature` with `tf_type` as the only argument.

If instead the feature's shape is given, it means the feature has a fixed length. Therefore, we use `FixedLenFeature`. A `FixedLenFeature` can also have a default value, which will be specified in `feature_config` if the key `'default_value'` is present.

In the `else` block, set `default_value` equal to the value in `feature_config` mapped to by `'default_value'`. If the key is not present, set `default_value` equal to `None`.

Then set `feature` equal to `tf.FixedLenFeature` with `shape`, `tf_type`, and `default_value` as the arguments (in that order).

Outside the `if...else` block, return `feature`.

```
import tensorflow as tf

def make_feature_config(shape, tf_type, feature_config):
    # CODE HERE
    pass

def create_example_spec(config):
    example_spec = {}
    for feature_name, feature_config in config.items():
        if feature_config['type'] == 'int':
            tf_type = tf.int64
        elif feature_config['type'] == 'float':
            tf_type = tf.float32
        else:
            tf_type = tf.string
        shape = feature_config['shape']
        feature = make_feature_config(shape, tf_type, feature_config)
        example_spec[feature_name] = feature
    return example_spec
```

Parsing

Parse data from serialized protocol buffers.

Chapter Goals:

- Learn how to parse a serialized protocol buffer
- Understand how feature data is represented in TensorFlow
- Implement a function that parses a serialized protocol buffer

A. Feature parsing

After creating an Example spec, it can be used to parse serialized protocol buffers that are read from a TFRecords file. Specifically, we use the Example spec as an argument to the `tf.parse_single_example` function, which converts a serialized protocol buffer into a usable feature dictionary.

```
import tensorflow as tf

print(example_spec)
print(repr(ex))

parsed = tf.parse_single_example(
    ex.SerializeToString(), example_spec)
print(repr(parsed))
```

You'll notice that the output of `tf.parse_single_example` is a dictionary mapping feature names to either a `tf.Tensor` or a `tf.SparseTensor`. Each `FixedLenFeature` is converted to a `tf.Tensor` and each `VarLenFeature` is converted to a `tf.SparseTensor`.

A `tf.Tensor` is basically TensorFlow's version of NumPy arrays, meaning it is a container for feature data with a fixed shape. `tf.SparseTensor` is used to represent data that may have many missing or empty values, making it useful for variable-length features.

In upcoming chapters, we'll discuss how we can combine the `tf.Tensor` and `tf.SparseTensor` values of the parsed dictionary to create an input layer for a TensorFlow model.

B. Shapes: `()` vs. `1`

In the previous chapter, we brought up how using `()` for the shape of a `FixedLenFeature` is different from using `1`.

Using `()` (or `[]`) corresponds to a single data value, while using `1` (represented as `(1,)` in `tf.Tensor`) corresponds to a list containing a single data value.

Time to Code!

In this chapter you'll complete the `parse_example` function, which parses data from a serialized Example object.

Using the input Example spec, `example_spec`, we'll parse the serialized protocol buffer, `example_bytes`.

Set `parsed_features` equal to `tf.parse_single_example` with `example_bytes` as the first argument and `example_spec` as the second argument.

After parsing the serialized protocol buffer into a dictionary, we may only want to return certain features. The features we return depend on the value of `output_features`.

If `output_features` is `None` (the default), we'll return the entire parsed dictionary. Otherwise, we only return the key-value pairs if the key is in the `output_features` list.

If `output_features` is not `None`, set `parsed_features` to a dictionary containing only the key-value pairs for keys that are listed in `output_features`.

Return `parsed_features`.

```
import tensorflow as tf
```

```
def parse_example(example_bytes, example_spec, output_features=None):
    # CODE HERE
    pass
```

Dataset

Learn how datasets are represented in TensorFlow's input pipeline.

Chapter Goals:

- Learn how to create a dataset in TensorFlow
- Implement a function that creates a dataset from NumPy data

A. Input pipeline

In TensorFlow, the input pipeline for executing a machine learning model is represented by the `Dataset` class (which we'll refer to as simply a dataset). A dataset can be created for a variety of input values, from NumPy arrays to protocol buffers. The most basic way to create a dataset is with the `tf.data.Dataset.from_tensor_slices` function.

```
import numpy as np
import tensorflow as tf

data = np.array([[  1. ,   2.1],
       [  2. ,   3. ],
       [  8.1, -10. ]])

d1 = tf.data.Dataset.from_tensor_slices(data)

print(d1)
```

Using from_tensor_slices to create a dataset from a NumPy array.

In the example, `d1` is a dataset containing the data from `data`. The dataset consists of three observations, with each observation being a row in `data`. Since each row of `data` has two columns, the observations in `d1` have shape `(2,)`.

We can also create datasets from tuple inputs. This is useful when we want to create a dataset from both feature data and labels for each data observation.

```
import numpy as np
import tensorflow as tf

data = np.array([[1. , 2. , 3. ],
       [1.1, 0. , 8. ]])

labels = np.array([1, 0])

d2 = tf.data.Dataset.from_tensor_slices((data, labels))

print(d2)
```

Using from_tensor_slices to create a dataset from a tuple of NumPy arrays.

In the example, `d2` is a dataset containing the data from `data` and the observation labels from `labels`. There are two total observations, and each observation has shape `(3,)`, since `data` has three columns.

B. Image file dataset

The `from_tensor_slices` function is not limited to just taking NumPy arrays as input. For example, we can use it to create a dataset of file names. A popular application of this is creating a dataset for image files.

```
import numpy as np
import tensorflow as tf

filenames = ['img1.jpg', 'img2.jpg']
img_d1 = tf.data.Dataset.from_tensor_slices(filenames)
print(img_d1)

labels = np.array([1, 0])
img_d2 = tf.data.Dataset.from_tensor_slices((filenames, labels))
print(img_d2)
```

Using from_tensor_slices to create a dataset for image file names.

In the example, `img_d1` represents a dataset for the input file names, while `img_d2` also has a label for each image file. Note that each dataset observation is a filename, rather than the actual file contents. For more information on

processing image files to retrieve the byte data, see the **Image Recognition** course on Educative.

C. Specialized datasets

Apart from the `from_tensor_slices` function, we can also use `TFRecordDataset` and `TextLineDataset` to create specialized datasets for protocol buffers and text data, respectively.

```
import numpy as np
import tensorflow as tf

records_files = ['one.tfrecords', 'two.tfrecords']
d1 = tf.data.TFRecordDataset(records_files)
print(d1)

txt_files = ['lines.txt']
d2 = tf.data.TextLineDataset(txt_files)
print(d2)
```

Using TFRecordDataset and TextLineDataset to create specialized datasets.

The `TFRecordDataset` takes in a list of TFRecords files and creates a dataset where each observation is an individual serialized protocol buffer. In the example, `d1` contains the serialized protocol buffers from `'one.tfrecords'` and `'two.tfrecords'`.

The `TextLineDataset` takes in a list of text files and creates a dataset where each observation is a separate line from the text files. In the example, `d2` contains the lines from `'lines.txt'`.

Mapping

Transform each individual observation in a dataset through mapping.

Chapter Goals:

- Learn how to map a function onto each observation of a dataset
- Implement a function that creates a dataset of serialized protocol buffers and parses each observation

A. Mapping function

After initially creating a dataset from NumPy arrays or files, we oftentimes want to apply changes to make the dataset observations more useful. For example, we might create a dataset from heights measured in inches, but we want to train a model on the heights in centimeters. We can convert each observation to the desired format by using the `map` function.

```
import numpy as np
import tensorflow as tf

data = np.array([65.2, 70. ])
d1 = tf.data.Dataset.from_tensor_slices(data)
d2 = d1.map(lambda x:x * 2.54)
print(d2)
```

Using the map function to convert a dataset of heights from inches to cm.

In the example above, `d1` is a dataset containing the height values from `data`, measured in inches. We use `map` to apply a function onto each observation of `d1`. The mapping function (represented by the lambda input to `map`) multiplies each observation value by 2.54 (the inch-centimeter conversion).

The output of `map`, which is `d2` in the example, is the resulting dataset containing the mapped observation values. In this case, the values of `d2` will be 165.608 and 182.88.

When a dataset is created from a tuple, the input function for `map` must take in a tuple as its argument.

```
import numpy as np
import tensorflow as tf

data1 = np.array([[1.2, 2.2],
       [7.3, 0. ]])
data2 = np.array([0.1, 1.1])
d1 = tf.data.Dataset.from_tensor_slices((data1, data2))
d2 = d1.map(lambda x,y:x + y)
print(d2)
```

Using the map function to convert a dataset of tuple observations.

B. Wrapper functions

One thing to note about `map` is that its input function must only take in a single argument, representing an individual dataset observation. However, we may want to use a multi-argument function as the input to `map`. In this case, we can use a *wrapper* to ensure that the input function is in the correct format.

```
import numpy as np
import tensorflow as tf

def f(a, b):
       return a - b
data1 = np.array([[4.3, 2.7],
       [1.3, 1. ]])
data2 = np.array([0.2, 0.5])
d1 = tf.data.Dataset.from_tensor_slices(data1)
d2 = d1.map(lambda x:f(x, data2))
print(d2)
```

Using the map function with a wrapper function around function f.

In the example above, `f` is an external function that subtracts its second argument from its first argument. To use `f` as the mapping function for `d1` (with `data2` as the second argument), we create a wrapper function for `f`.

represented by the lambda input to `map`.

The wrapper function takes in a single argument, `x`, so it meets the criteria as an input to `map`. It then uses `x` as the first argument to `f`, while using `data2` as the second argument.

Time to Code!

In this chapter you'll be completing the `dataset_from_examples` function, which maps the `parse_example` function from chapter 5 onto a `TFRecordDataset`.

The first thing we'll do is create the Example spec that's used for parsing, by using the `create_example_spec` function from chapter 4.

Set `example_spec` equal to `create_example_spec` applied with `config` as the only argument.

Next, we create a dataset from the TFRecords files given by `filenames`.

Set `dataset` equal to `tf.data.TFRecordsDataset` initialized with `filenames`.

The `dataset` we created contains serialized protocol buffers for each observation. To parse each serialized protocol buffer, we need to map the `parse_example` function from chapter 5 onto each observation of the dataset.

Since the input function for `map` can only take in a single argument, we'll create a lambda wrapper around the `parse_example` function.

Set `wrapper` equal to a lambda function whose input argument is named `example`. The lambda function should return `parse_example` applied with `example`, `example_spec`, and `output_features` as the first, second, and third arguments.

Finally, we can apply the `map` function onto the dataset and return the output.

Set `dataset` equal to `dataset.map` applied with `wrapper` as the input function. Then return `dataset`.

```
import tensorflow as tf

def create_example_spec(config):
    example_spec = {}
```

```
    example_spec = {}
    for feature_name, feature_config in config.items():
        if feature_config['type'] == 'int':
            tf_type = tf.int64
        elif feature_config['type'] == 'float':
            tf_type = tf.float32
        else:
            tf_type = tf.string
        shape = feature_config['shape']
        if shape is None:
            feature = tf.VarLenFeature(tf_type)
        else:
            default_value = feature_config.get('default_value', None)
            feature = tf.FixedLenFeature(shape, tf_type, default_value)
        example_spec[feature_name] = feature
    return example_spec

def parse_example(example_bytes, example_spec, output_features=None):
    parsed_features = tf.parse_single_example(example_bytes, example_spec)
    if output_features is not None:
        parsed_features = {k: parsed_features[k] for k in output_features}
    return parsed_features

# Map the parse_example function onto a TFRecord Dataset
def dataset_from_examples(filenames, config, output_features=None):
    # CODE HERE
    pass
```

Configuration

Configure a dataset with shuffling, repetitions, and batch size.

Chapter Goals:

- Learn how to configure a dataset
- Implement a function that shuffles, repeats, and batches an input dataset

A. Shuffling

When using a dataset to train a machine learning model, there are certain things we need to do to properly configure the dataset. When we first create a dataset from NumPy arrays or files, the observations may be ordered in a particular way. For example, many data files will sort the data observations by some particular feature, like a person's name or year.

While systematic ordering of data files makes it easier for humans to look over the data, it actually hinders the training of a machine learning model. The model will learn to make predictions based on the ordering of the observations rather than the observations themselves, which is not what we want our model to do. To avoid this, we need to randomly shuffle our dataset prior to training a model, which is done with the `shuffle` function.

```
import numpy as np
import tensorflow as tf

data = np.random.uniform(-100, 100, (1000, 5))
original = tf.data.Dataset.from_tensor_slices(data)

shuffled1 = original.shuffle(100)
print(shuffled1)

shuffled2 = original.shuffle(len(data))
print(shuffled2)
```

Using the shuffle function to randomly shuffle a dataset.

In the example, `original` contains 1000 observations, with each observation consisting of 5 floating point numbers. The datasets `shuffled1` and `shuffled2` are the results of randomly shuffling `original` using different *buffer sizes*.

The buffer size (specified by the required argument of `shuffle`) essentially dictates how random our shuffling is. The larger the buffer size, the more uniformly random the shuffling will be.

If the buffer size is 1, then no shuffling occurs. If the buffer size is $\geq$ the number of total observations (the case for `shuffled2`), then the shuffling is uniformly random across the entire dataset. When the buffer size is somewhere in between (the case for `shuffled1`), shuffling will still occur but it will not be uniformly random for all the observations.

In most cases, uniform shuffling is the best option, since that's the safest bet to avoiding any systematic ordering of the initial data observations. However, for datasets that take up a lot of memory, it may be useful to initially shuffle the entire dataset once, then use a smaller buffer size for faster training.

B. Epochs

Aside from randomly shuffling, we also need to configure a dataset so that training can be done for multiple *epochs*. An epoch refers to a single training run over the entire dataset. We normally need to go through several epochs of a dataset before the model is finished training.

Without any configuration, a dataset can only be used to train a model for one epoch. If we try to train for more than one epoch, we get back an `OutOfRangeError`. To fix this, we use the `repeat` function to specify the number of epochs we can run a dataset.

```
import numpy as np
import tensorflow as tf

data = np.random.uniform(-100, 100, (1000, 5))
original = tf.data.Dataset.from_tensor_slices(data)

repeat1 = original.repeat(1)
print(repeat1)

repeat2 = original.repeat(100)
print(repeat2)
```

```
print(repeat1)

repeat3 = original.repeat()
print(repeat3)
```

Using the repeat function to configure a dataset for multiple epochs.

The `repeat` function takes in a single (optional) argument, which specifies how many epochs we can run the dataset for.

The `repeat1` dataset is explicitly configured for one epoch, while `repeat2` is configured for 100 epochs. This means we can use `repeat1` for one full run through of `original` and `repeat2` for 100 full run throughs of `original`, before getting back an `OutOfRangeError`.

For `repeat3`, we didn't pass in any argument. This is the same as passing in `None`, which is the default value for the optional argument. It means that `repeat3` can be run indefinitely, and will never raise an `OutOfRangeError`.

C. Batching

The default number of data observations we use for each dataset iteration is one. This can be incredibly slow when training or evaluating very large datasets, which can have millions of observations. Luckily, we can explicitly set the number of observations per iteration (i.e. the *batch size*) using the `batch` function.

```
import numpy as np
import tensorflow as tf

data = np.random.uniform(-100, 100, (1000, 5))
original = tf.data.Dataset.from_tensor_slices(data)

batch1 = original.batch(1)
print(batch1)

batch2 = original.batch(100)
print(batch2)
```

Using the batch function to set explicit batch sizes for each iteration of observations. The ? in the shape of the batched datasets is just a placeholder for the batch size.

The `batch` function takes in a required argument, representing the batch size per iteration. In the example, `batch1` will use one data observation per iteration while `batch2` will use 100. So at each iteration, the shape of the data for `batch1` will be `(1, 5)`, while the shape of the data for `batch2` will be `(100, 5)`.

In general, it is a good idea to explicitly set the batch size using `batch`, even if the batch size is just 1. Without setting the batch size, the shape of the data at each iteration is 1-D (in the example, it is `(5,)`). This can potentially cause issues, since a single observation might be interpreted as a batch (e.g. a shape of `(5,)` could be viewed as a batch of 5 observations).

The optimal batch size will differ depending on the size of the dataset and the problem itself. The batch size can also change at different points of the training. Initially, you may want a larger batch size so the model can train faster in its early stages, then gradually decrease the batch size as the model gets closer to converging so as to not overshoot the convergence point.

Dataset Iteration

Iterate through a dataset to extract individual data observations.

Chapter Goals:

- Learn how to iterate through a dataset and extract values from data observations
- Implement a function that iterates through a NumPy-based dataset and extracts the feature data

A. Iterator

The previous few chapters focused on creating and configuring datasets. In this chapter, we'll discuss how to iterate through a dataset and extract the data.

To iterate through a dataset, we need to create an `Iterator` object. There are a few different ways to create an Iterator, but we'll focus on the simplest and most commonly used method, which is the `make_one_shot_iterator` function.

```
import numpy as np
import tensorflow as tf

data = np.array([[1., 2.],
       [3., 4.]])
dataset = tf.data.Dataset.from_tensor_slices(data)
dataset = dataset.batch(1)

it = dataset.make_one_shot_iterator()
next_elem = it.get_next()
print(next_elem)

added = next_elem + 1
print(added)
```

In the example, `it` represents an Iterator for `dataset`. The `get_next` function returns something we'll refer to as the next-element tensor.

The next-element tensor represents the batched data observation(s) at each iteration through the dataset. We can even apply operations or transformations to the next-element tensor. In the example above, we added 1 to each of the values in the data observation represented by `next_elem`.

B. Running the iteration

You'll notice that the next-element tensor is a `tf.Tensor` object. We use a `tf.Session` object to retrieve the values from a `tf.Tensor`.

`tf.Session` uses an important function called `run`, which allows us to extract the `tf.Tensor` values as NumPy data. For an in-depth look at `tf.Session` and the basics of TensorFlow execution, check out the Machine Learning for Software Engineers course.

```
import numpy as np
import tensorflow as tf

data = np.array([[1., 2.],
       [3., 4.]])
dataset = tf.data.Dataset.from_tensor_slices(data)
dataset = dataset.batch(1)

it = dataset.make_one_shot_iterator()
next_elem = it.get_next()
added = next_elem + 1

sess = tf.Session()
print('First elem in batch: {}'.format(
    repr(sess.run(added))))
print('Second elem in batch: {}'.format(
    repr(sess.run(added))))
print()  # Newline
try:
    sess.run(added)  # OutOfRangeError
except tf.errors.OutOfRangeError:
    # New session
    with tf.Session() as sess:
        for i in range(2):
            print(repr(sess.run(added)))
```

Using tf.Session to extract values from the added variable.

Similar to File I/O in Python, we can create a `tf.Session` with or without the

`with` keyword. However, the `with` keyword lets us define all our computation

within the scope of the `tf.Session` object, so we don't have to manually close it to free its resources.

In the example, the ith time we call `sess.run` on `added` will return the ith observation from the dataset. Since we used a $+1$ transformation to obtain `added` from `next_elem`, each observation's values are incremented by 1.

Notice that if we call `sess.run` three consecutive times within the same `tf.Session` object scope, an `OutOfRangeError` is raised on the third call. This is because the dataset only contains two data observations, and we didn't use the `repeat` function to increase the number of epochs we can iterate through.

C. Configured dataset

The dataset used in the previous two examples was somewhat simplistic, and only intended to showcase the basics of the iteration process. For a more complex example, we'll iterate through a dataset configured with `shuffle`, `repeat`, and `batch`.

```
import numpy as np
import tensorflow as tf

data = np.array([
  [1., 2.],
  [3., 4.],
  [5., 6.],
  [7., 8.],
  [0., 9.],
  [0., 0.]])

dataset = tf.data.Dataset.from_tensor_slices(data)
dataset = dataset.shuffle(6)
dataset = dataset.repeat()
dataset = dataset.batch(2)
it = dataset.make_one_shot_iterator()
next_elem = it.get_next()
with tf.Session() as sess:
  for i in range(4):
    print('Element {}: {}'.format(
      i + 1, repr(sess.run(next_elem))))
```

Iterating through a fully configured dataset.

The first thing to notice is that, despite `dataset` having only six data observations, we were able to iterate through eight observations because we used the `repeat` function. In fact, since we used `repeat` with its default argument setting, we could continuously iterate through the dataset without raising an `OutOfRangeError`.

Since we set the batch size to 2 using `batch`, each iteration returned two data observations rather than 1. Furthermore, you'll notice that the observations appear in a random order due to `shuffle`. However, we still saw all the data observations within the first epoch (i.e. first three iterations), because the shuffling occurs on a per-epoch basis.

Feature Columns

Learn about feature columns and how they're used to extract data features.

Chapter Goals:

- Learn about feature columns and how they're used
- Implement a function that creates a list of feature columns

A. Overview

Before we get into using a dataset of parsed protocol buffers, we need to first discuss *feature columns*. In TensorFlow, a feature column is how we specify what kind of data a feature contains. In this chapter, we'll focus on the two most common types of feature data: numeric and categorical data.

Feature columns are incredibly useful for converting raw data into an input layer for a machine learning model. Once we have a list of feature columns, we can use them to combine `tf.Tensor` and `tf.SparseTensor` feature data into a single input layer. We'll discuss more of this in the next chapter.

B. Numeric features

For numeric features, we create a feature column using `tf.feature_column.numeric_feature`. The function takes in the feature name as a required argument.

```
import tensorflow as tf

nc = tf.feature_column.numeric_column(
  'GPA', shape=5, dtype=tf.float32)

print(nc)
```

In the example above, `nc` represents a numeric feature column for the feature called `'GPA'`. We used the `shape` keyword argument to specify that the

feature called `GPA`. We used the `shape` keyword argument to specify that the feature must be 1-D and contain 5 elements. We also set the feature's datatype to `tf.float32`.

Other less commonly used keyword arguments for the function are `default_value` and `normalizer_fn`.

The `default_value` keyword argument sets the default value for the feature column if the feature is missing from the protocol buffer. If `None` (which is the default), it means that the specified feature must be present in all entries in order for us to use the numeric column without error.

The `normalizer_fn` keyword argument takes in a function that normalizes the feature data. This would be used in a similar way to how `map` is used for datasets.

C. Categorical features

To create a feature column for categorical data, we need a *vocabulary* for the data. A vocabulary simply refers to the list of possible categories for the data. There are two functions we use to create categorical feature columns:

- `tf.feature_column.categorical_column_with_vocabulary_list`
- `tf.feature_column.categorical_column_with_vocabulary_file`

From this point on, we'll refer to these functions without their `tf.feature_column` prefix, for brevity.

```
import tensorflow as tf

cc1 = tf.feature_column.categorical_column_with_vocabulary_list(
  'name', ['a', 'b', 'c'])

cc2 = tf.feature_column.categorical_column_with_vocabulary_list(
  'name', [1, 2, 3])

cc3 = tf.feature_column.categorical_column_with_vocabulary_list(
  'name', ['a', 'b', 'NA'], default_value=2)

cc4 = tf.feature_column.categorical_column_with_vocabulary_list(
  'name', ['a', 'b', 'c'], num_oov_buckets=2)
```

The `categorical_column_with_vocabulary_list` function takes in two required arguments: the feature name and the list of categories (vocabulary).

The `dtype` keyword argument specifies the datatype for the feature column. If `dtype` is not passed in, the datatype is inferred from the vocabulary list. In our example, `cc1` has type `tf.string` and `cc2` has type `tf.int32`.

The `default_value` argument specifies a non-negative default index for an OOV (out-of-vocabulary) category. In our example, each OOV value for `cc3` would be mapped to the `'NA'` category.

The `num_oov_buckets` argument specifies the number of OOV "dummy" categories to create. These "dummy" values will be filled when we actually run our TensorFlow computation graph. In our example, we create two "dummy" categories in `cc4` for placing OOV values into.

Note that `default_value` and `num_oov_buckets` keyword arguments cannot be used together. `default_value` assigns an already existing category to catch OOV values, while `num_oov_buckets` creates new categories to put OOV values in.

task.py

vocab.txt

```
import tensorflow as tf

cc1 = tf.feature_column.categorical_column_with_vocabulary_file(
  'name', 'vocab.txt')

cc2 = tf.feature_column.categorical_column_with_vocabulary_file(
  'name', 'vocab.txt', vocabulary_size=4)
```

The `categorical_column_with_vocabulary_file` function takes in two required arguments: the feature name and the file containing the vocabulary. The vocabulary file must contain a single category per line.

The `vocabulary_size` keyword argument specifies how many of the

The `vocabulary_size` keyword argument specifies how many of the vocabulary categories to use. For `cc2`, we specified that the feature column

only use the first 4 categories of the 11 categories listed in the file. When we don't set this argument, the function will infer the vocabulary size based on the number of lines in the file.

The remaining keyword arguments are identical to the ones in `categorical_column_with_vocabulary_list`.

D. Indicator wrapping

Unlike numeric feature columns, categorical feature columns cannot be used directly to create the input layer for a machine learning model. In order to actually use a categorical feature column, we need to convert the categorical data into numeric values. One of the most common ways to do this is with the `tf.feature_column.indicator_column` function.

task.py

vocab.txt

```
import tensorflow as tf

cc = tf.feature_column.categorical_column_with_vocabulary_file(
  'name', 'vocab.txt')

ic = tf.feature_column.indicator_column(cc)
print(ic)
```

The indicator column wrapper converts the categorical data into a count vector. The count vector gives a count of how many times each vocabulary category appears in the feature data. You'll see an explicit example of count vectors in the next chapter.

Time to Code!

In this chapter you'll be completing the `create_feature_columns` function, which creates a list of feature columns for a variety of features. The feature configurations are given by the input dictionary, `config`.

Specifically, you'll be creating three helper functions: `create_list_column`, `create_file_column`, and `create_numeric_column`. These functions are used to create the necessary feature columns for `create_feature_columns`:

```
import tensorflow as tf

def create_feature_columns(config, example_spec, output_features=None):
    if output_features is None:
        output_features = config.keys()
    feature_columns = []
    for feature_name in output_features:
        dtype = example_spec[feature_name].dtype
        feature_config = config[feature_name]
        # HELPER FUNCTIONS USED
        if 'vocab_list' in feature_config:
            feature_col = create_list_column(feature_name, feature_config, dtype)
        elif 'vocab_file' in feature_config:
            feature_col = create_file_column(feature_name, feature_config, dtype)
        else:
            feature_col = create_numeric_column(feature_name, feature_config, dtype)
        feature_columns.append(feature_col)
    return feature_columns
```

The `create_list_column` function creates a categorical feature from a vocabulary list. To do this, we use the `tf.feature_column.categorical_column_with_vocabulary_list` function.

Set `vocab_feature_col` equal to the specified function applied with `feature_name` and `feature_config['vocab_list']` as the first two arguments, and `dtype=dtype` for the keyword argument.

Since `vocab_feature_col` is a categorical column, we need to wrap it with an indicator column.

Set `feature_col` equal to `tf.feature_column.indicator_column` applied with `vocab_feature_col` as the only argument. Then return `feature_col`.

```
import tensorflow as tf

def create_list_column(feature_name, feature_config, dtype):
    # CODE HERE
    pass
```

The `create_file_column` function creates a categorical feature from a vocabulary file. To do this, we use the `tf.feature_column.categorical_column_with_vocabulary_file` function.

Set `vocab_feature_col` equal to the specified function applied with `feature_name` and `feature_config['vocab_file']` as the first two arguments, and `dtype=dtype` for the keyword argument.

Since `vocab_feature_col` is a categorical column, we need to wrap it with an indicator column.

Set `feature_col` equal to `tf.feature_column.indicator_column` applied with `vocab_feature_col` as the only argument. Then return `feature_col`.

task.py

majors.txt

```
import tensorflow as tf

def create_file_column(feature_name, feature_config, dtype):
    # CODE HERE
    pass
```

The `create_numeric_column` function creates a numeric feature. To do this, we use the `tf.feature_column.numeric_column` function.

Set `feature_col` equal to the specified function applied with `feature_name` as the first argument, `feature_config['shape']` for the `shape` keyword argument, and `dtype` for the `dtype` keyword argument. Then return `feature_col`.

```
import tensorflow as tf

def create_numeric_column(feature_name, feature_config, dtype):
    # CODE HERE
    pass
```

Input Layer

Combine data features into an input layer for a neural network.

Chapter Goals:

- Learn how to use feature columns to create an input layer for a machine learning model
- Implement a function that iterates through a protocol buffer dataset and creates a list of input layers

A. Combining features

If you recall, a parsed protocol buffer is a dictionary that maps feature names to either a `tf.Tensor` or `tf.SparseTensor` object. Using the parsed protocol buffer and corresponding feature columns, we can combine the feature values into an input layer for a machine learning model. To do this, we use the `tf.feature_column.input_layer` function.

task.py

majors.txt

```
import numpy as np
import tensorflow as tf

# Get a parsed TFRecordDataset
dataset = get_dataset()

it = dataset.make_one_shot_iterator()
next_elem = it.get_next()
# Get feature columns for the dataset
feature_columns = get_feature_columns()
for i in range(len(feature_columns)):
    print(feature_columns[i])
    print() # Newline
inputs = tf.feature_column.input_layer(next_elem, feature_columns)
print(inputs)
```

In the example above, the `inputs` tensor represents an input layer created from a batch of parsed protocol buffers. Notice that the second dimension of the tensor's shape is 21. This means that the features specified in `feature_columns` have a combined value length of 21.

B. Extracting data

We use `tf.Session` to extract values from the input layer tensor. The process of iterating through the dataset and extracting data is nearly identical to the one described in chapter 9. The only difference is that, when using categorical feature columns, we need to set up a tables initializer with the `tf.tables_initializer` operation.

task.py

majors.txt

```
import numpy as np
import tensorflow as tf

# Get a parsed TFRecordDataset
dataset = get_dataset()

it = dataset.make_one_shot_iterator()
next_elem = it.get_next()
# Get feature columns for the dataset
feature_columns = get_feature_columns()
inputs = tf.feature_column.input_layer(next_elem, feature_columns)

table_init = tf.tables_initializer()
with tf.Session() as sess:
    sess.run(table_init) # Initialize vocab table
    for i in range(2):
        print(repr(sess.run(inputs)))
```

This code will extract values from the input layer tensor

Since there are vocabulary features in `feature_columns`, we need to use the table initializer in the example above. We call `sess.run` on `table_init` to perform the table initialization.

Quiz

1

What is the input pipeline for a machine learning model?

COMPLETED 0%

1 of 4

Introduction

An overview of the development process for machine learning models.

In this section you will be learning about the model execution process in TensorFlow. This involves model training, evaluation, and making predictions. The techniques and APIs introduced in the upcoming chapters are used by top industry professionals to easily create organized and efficient machine learning models.

The first model you create is going to be a classification model. Classification models take data observations and label them with a certain category. For example, classification models can used for tasks like fraud detection, where something is categorized as either fraud or legitimate.

The second model you create is going to be a regression model. Regression models take data observations and output a number. which which can be used for predicting sales with this course's supermarket dataset.

A. Creating a model

Machine learning models all follow the same format; they require an input layer, apply some computations, and produce an output. In the following chapters, we will cover this entire process using a generic multilayer perceptron (MLP) neural network.

B. Development process

The development process for a machine learning model involves continuous training and evaluation. Many machine learning models take an incredibly long time to train. Some take days, or even weeks, to finish training.

Because of this long training time, it is useful to periodically save the current model state while training. This allows us to stop training if we need to and resume training from the same spot at a later date. It also lets us restore previous versions of a trained model, in case something goes wrong or we

overtrain.

Being able to save and restore models in a streamlined fashion is one of TensorFlow's best features. Another strength of the TensorFlow library is the ability to easily log certain values, such as a model's accuracy or loss. This is important in making sure we notice any strange occurrences during training.

TensorBoard

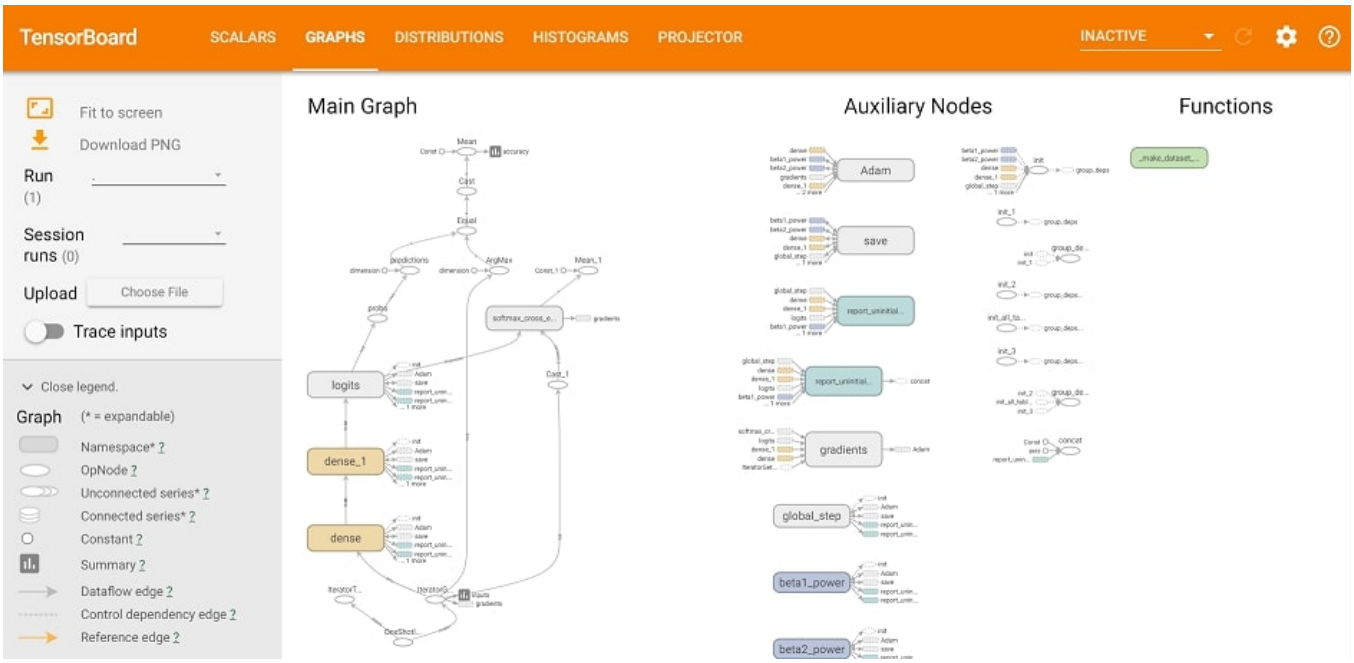
Discover how TensorBoard can be used to visualize machine learning.

Chapter Goals:

- Learn about TensorBoard and how to track the progression of training values
- Specify training values to be shown in TensorBoard

A. Training visualizations

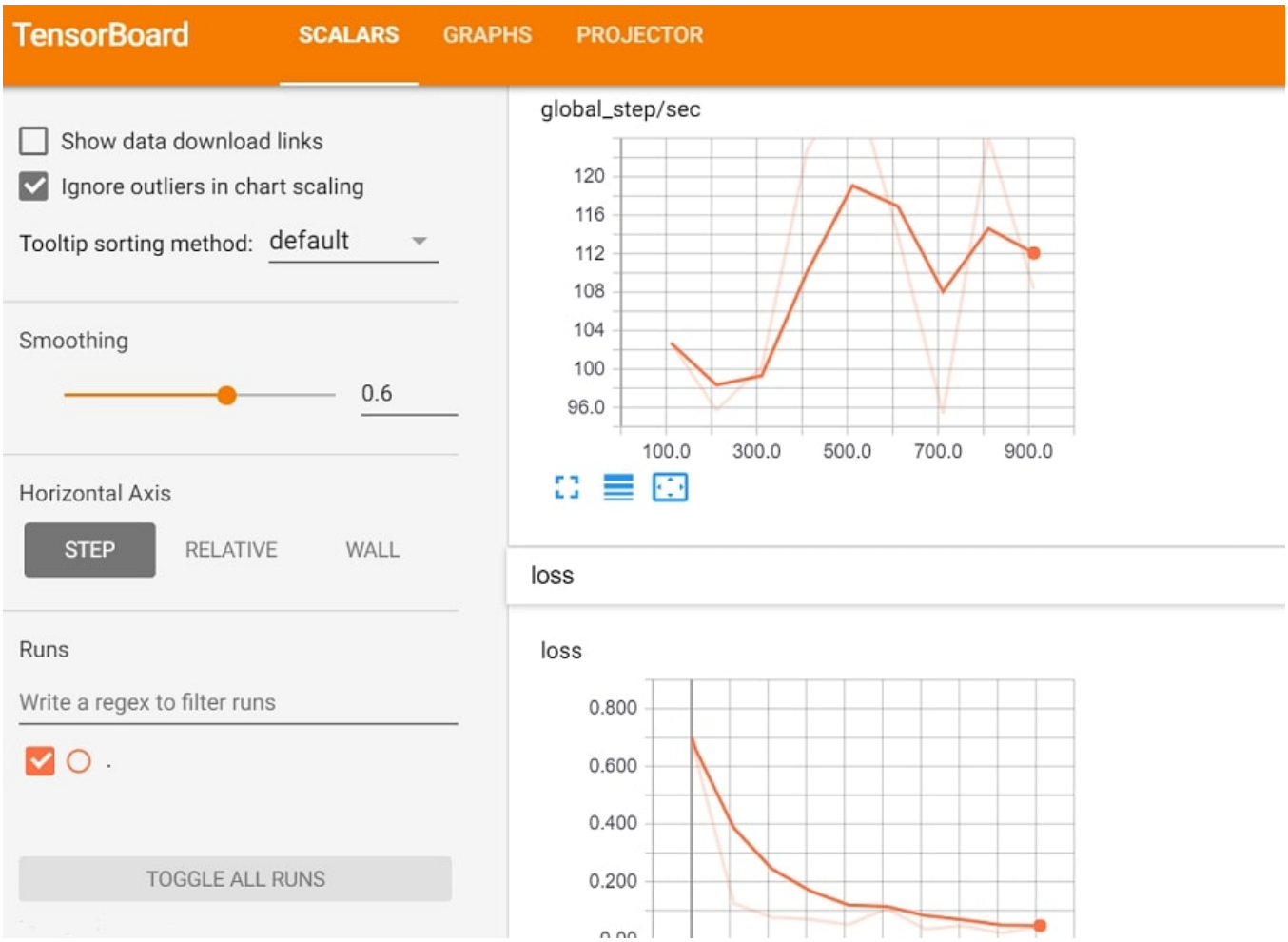
When training a complex neural network, it is useful to have visualizations of the compuation graph and important values to make sure everything is correct. In TensorFlow, there is a tool known as TensorBoard, which lets us visualize all the important aspects of a model.



Computation graph structure of a neural network shown in TensorBoard.

TensorBoard works by reading in an *events file*, which contains all the model data we want visualized. When training a model, the events file is stored in the same directory as the model *checkpoint* (which we'll discuss in the next chapter). The events file automatically contains the computation graph structure, as well as the loss and training speed (in iterations per second).

structure, as well as the loss and training speed (in iterations per second).



Default events file values shown in TensorBoard. Note that "global_step/sec" represents the training iterations per second.

You can run TensorBoard from the command line with the built-in **tensorboard** module (which comes as part of the TensorFlow library). You just need to specify the directory containing the events file. TensorBoard will then be running in the browser at *http://localhost:6006*.

```
tensorboard --logdir=/home/ckpt_dir
```

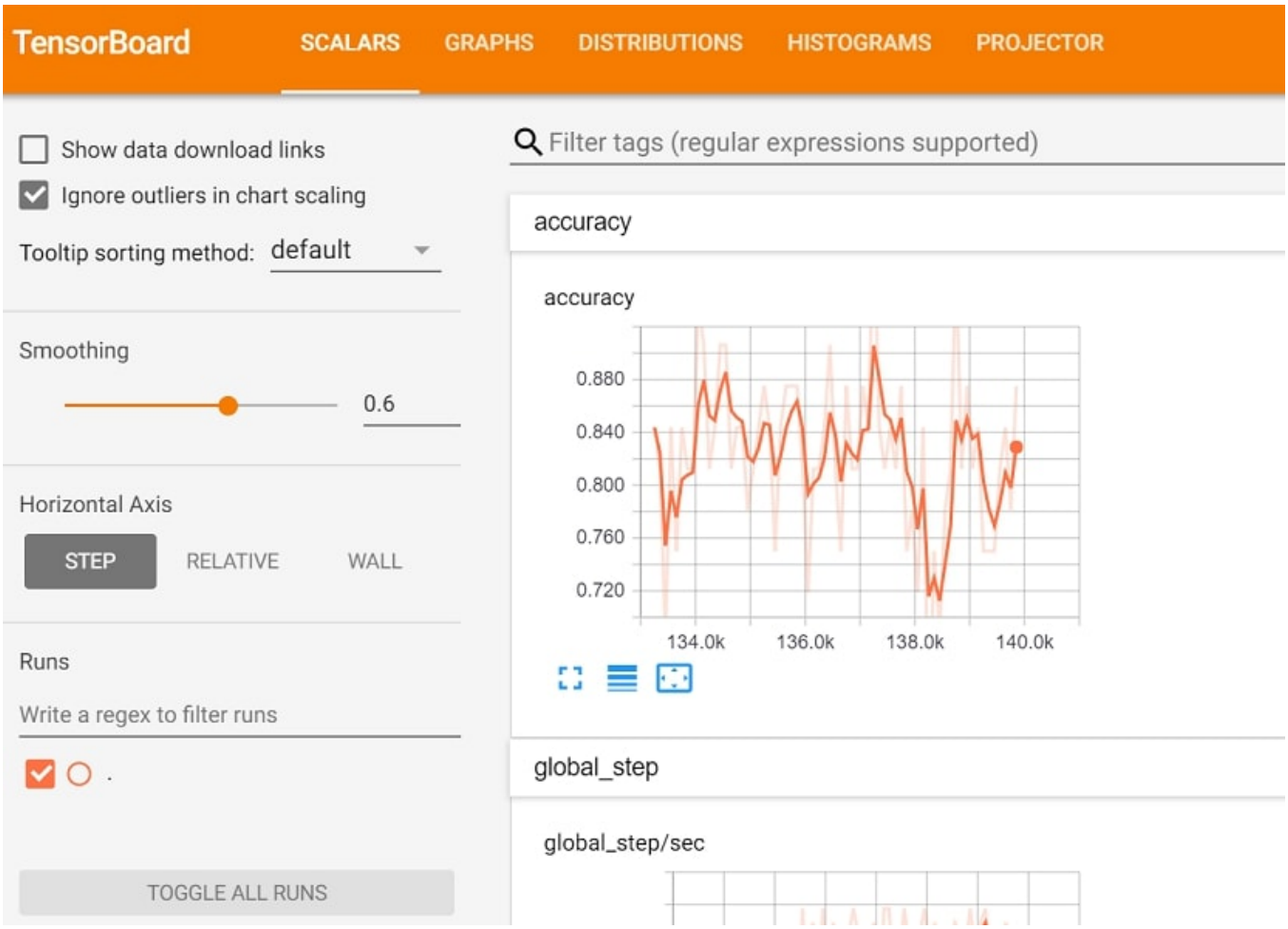
This command line statement runs TensorBoard. The events file is located in the /home/ckpt_dir directory.

B. Tracking values

Apart from the default events file values, we can also specify custom values to track in TensorBoard. To do this, we just need to call `tf.summary.scalar` in our code.

The function takes in two required arguments. The first argument is the label name for the visualization in TensorFlow. The second argument is the scalar

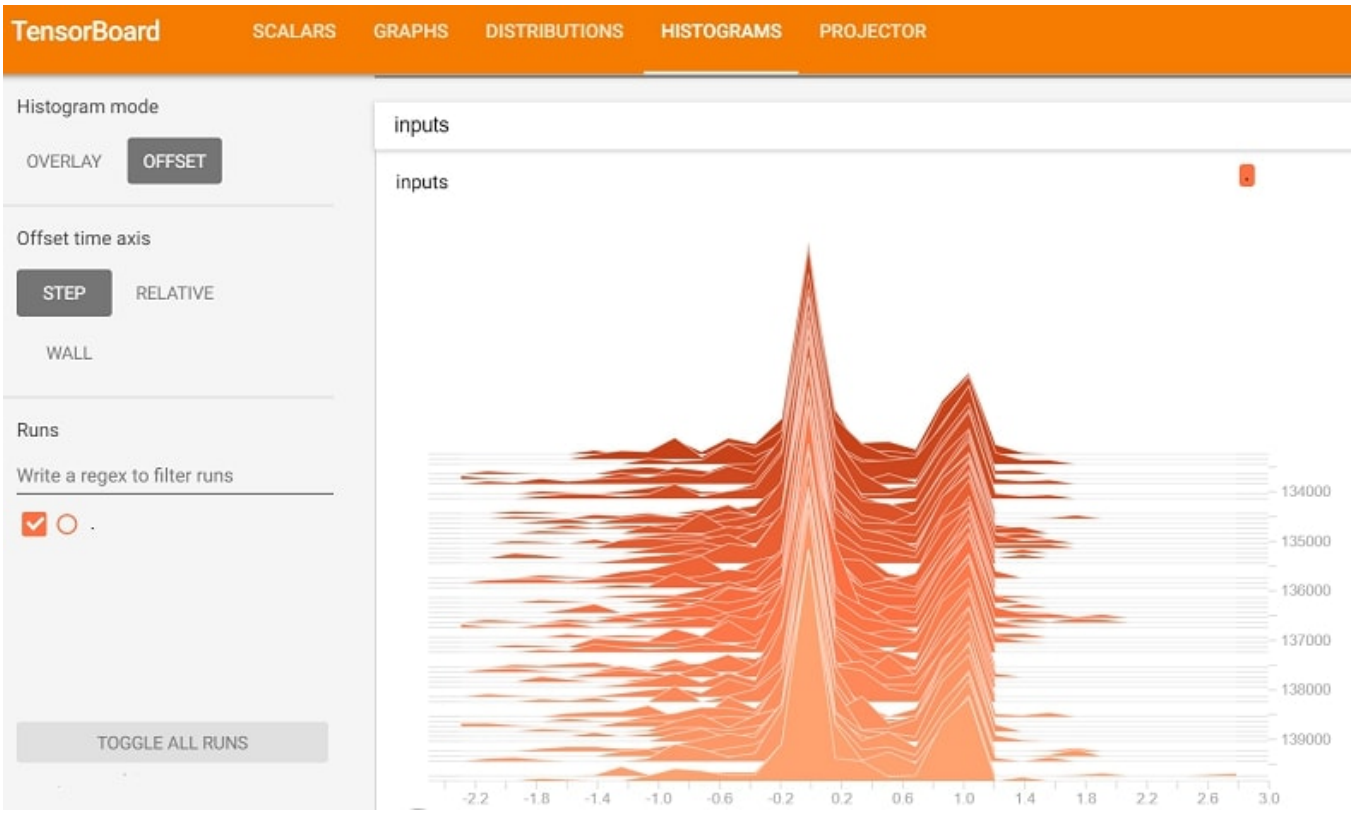
(i.e. single numeric value) tensor that will be visualized. The tensor's values will be plotted with respect to the training iterations.



Visualizing the model accuracy in TensorBoard. The plot shows the accuracy on the y-axis and training iteration on the x-axis.

We can also visualize the distribution of the values for a particular layer in the model. For example, we can view the distribution of the data in the input layer, or we can view the distribution of the weights in a particular hidden layer.

The function we use to visualize a distribution is `tf.summary.histogram`. This function takes in the same arguments as `tf.summary.scalar`.



Visualizing the distribution of the input layer in TensorBoard. The histogram shows the frequency of value appearances, with higher peaks representing more common values.

Time to Code!

The first seven chapters of this section of the course deals with creating a classification model in TensorFlow, represented by the `ClassificationModel` object. The function that you'll be working on this chapter and the next is `run_model_training`. This function will run training for a classification MLP model and log results to TensorBoard.

The function calls two helpers:

- `dataset_from_numpy`: creates a dataset from NumPy data
- `run_model_setup`: sets up the MLP model

Both functions are shown below.

```
import numpy as np
import tensorflow as tf

class ClassificationModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # See the "Efficient Data Processing Techniques" section for details
    def dataset_from_numpy(self, input_data, batch_size, labels=None, is_training=True, num_e
```

```
        dataset_input = input_data if labels is None else (input_data, labels)
        dataset = tf.data.Dataset.from_tensor_slices(dataset_input)
        if is_training:
            dataset = dataset.shuffle(len(input_data)).repeat(num_epochs)
        return dataset.batch(batch_size)

    # See the "Machine Learning for Software Engineers" course on Educative
    def run_model_setup(self, inputs, labels, hidden_layers, is_training, calculate_accuracy=
        layer = inputs
        for num_nodes in hidden_layers:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.probs = tf.nn.softmax(logits, name='probs')
        self.predictions = tf.argmax(
            self.probs, axis=-1, name='predictions')
        if calculate_accuracy:
            class_labels = tf.argmax(labels, axis=-1)
            is_correct = tf.equal(
                self.predictions, class_labels)
            is_correct_float = tf.cast(
                is_correct,
                tf.float32)
            self.accuracy = tf.reduce_mean(
                is_correct_float)
        if labels is not None:
            labels_float = tf.cast(
                labels, tf.float32)
            cross_entropy = tf.nn.softmax_cross_entropy_with_logits_v2(
                labels=labels_float,
                logits=logits)
            self.loss = tf.reduce_mean(
                cross_entropy)
        if is_training:
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)
```

In this chapter, you'll be creating another helper function called `add_to_tensorboard`, which adds metrics to log in TensorBoardg.

It's useful to keep track of how the model accuracy changes while training. Therefore, we'll want to make sure it's plotted in TensorBoard.

Call `tf.summary.scalar` with `'accuracy'` as the first argument and `self.accuracy` as the second argument.

We also want to store the distribution of the input layer (represented by `inputs`) in our TensorBoard. To visualize the distribution, we'll make sure to store it as a histogram.

Call `tf.summary.histogram` with `'inputs'` as the first argument and `inputs` as the second argument.

as the second argument.

```
import numpy as np
import tensorflow as tf

class ClassificationModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Adds metrics to TensorBoard
    def add_to_tensorboard(self, inputs):
        # CODE HERE
        pass

    # See the "Efficient Data Processing Techniques" section for details
    def dataset_from_numpy(self, input_data, batch_size, labels=None, is_training=True, num_e
        dataset_input = input_data if labels is None else (input_data, labels)
        dataset = tf.data.Dataset.from_tensor_slices(dataset_input)
        if is_training:
            dataset = dataset.shuffle(len(input_data)).repeat(num_epochs)
        return dataset.batch(batch_size)

    # See the "Machine Learning for Software Engineers" course on Educative
    def run_model_setup(self, inputs, labels, hidden_layers, is_training, calculate_accuracy=
        layer = inputs
        for num_nodes in hidden_layers:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.probs = tf.nn.softmax(logits, name='probs')
        self.predictions = tf.argmax(
            self.probs, axis=-1, name='predictions')
        if calculate_accuracy:
            class_labels = tf.argmax(labels, axis=-1)
            is_correct = tf.equal(
                self.predictions, class_labels)
            is_correct_float = tf.cast(
                is_correct,
                tf.float32)
            self.accuracy = tf.reduce_mean(
                is_correct_float)
        if labels is not None:
            labels_float = tf.cast(
                labels, tf.float32)
            cross_entropy = tf.nn.softmax_cross_entropy_with_logits_v2(
                labels=labels_float,
                logits=logits)
            self.loss = tf.reduce_mean(
                cross_entropy)
        if is_training:
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)

    # Run training of the classification model
    def run_model_training(self, input_data, labels, hidden_layers, batch_size, num_epochs, c
        self.global_step = tf.train.get_or_create_global_step()
        dataset = self.dataset_from_numpy(input_data, batch_size,
            labels=labels, num_epochs=num_epochs)
```

```
        iterator = dataset.make_one_shot_iterator()
        inputs, labels = iterator.get_next()


        self.run_model_setup(inputs, labels, hidden_layers, True)
        self.add_to_tensorboard(inputs)
```

Training

Learn about efficient and structured training in TensorFlow.

Chapter Goals:

- Understand how a `MonitoredTrainingSession` works
- Learn about saving checkpoints and tracking scalar values during training
- Train a machine learning model using a `MonitoredTrainingSession`

A. Logging values

While `tf.summary.scalar` lets us keep track of certain values in an events file for TensorBoard, it is also useful to directly log values to STDOUT during training. For instance, it is customary to log the loss and iteration count, so we can stop training if there is an issue.

```
python3 train_model.py
```

Running a script that trains a model for about 400 steps.

You'll notice each line of output is prepended by "INFO:tensorflow". This just means the logging level is set to INFO.

We log specific values while training using a `tf.train.LoggingTensorHook` object. The object is initialized with a dictionary mapping labels to scalar valued tensors. In our example, the labels we used were `'loss'` and `'step'`, for the loss and iteration count tensors, respectively. In the `run_model_training` function, `self.loss` represents the loss tensor and `self.global_step` represents the iteration count, also known as the training step.

To specify the logging frequency, we need to set exactly one of `every_n_iter`

To specify the logging frequency, we need to set exactly one of `every_n_iter` or `every_n_secs` as a keyword argument when initializing

`tf.train.LoggingTensorHook`. In the example above, we set `every_n_iter` to 100, so that logging is shown every 100 iterations.

We can also use `every_n_secs` to specify a time interval for displaying logged values.

B. Catching NaN

In addition to logging values during training, we also want to automatically stop training if the loss ever takes on an infinite value. In TensorFlow, similar to NumPy, an infinite value is represented by NaN.

```
python3 train_model_nan.py
```

NaN loss being caught by the tf.train.NanTensorHook. Training is stopped with an error message.

We use the `tf.train.NanTensorHook` to handle NaN loss. When initializing `tf.train.NanTensorHook`, we only need to pass in the loss tensor variable as a required argument.

C. Efficient training

We can use `tf.Session` and `tf.placeholder` to run model training. However, this is a relatively inefficient training method, and should really only be used to train small models or run tests/evaluation.

We've already shown how to replace `tf.placeholder` for the input data, by iterating through a TensorFlow dataset. We can also replace `tf.Session` with `tf.train.MonitoredTrainingSession`, which handles a lot of the training dirty work for us.

The `MonitoredTrainingSession` will initialize all the necessary variables in the computation graph and log specified scalar values. We can run training within the scope of a `MonitoredTrainingSession` object by using the `with` keyword.

Below we show how to use a `MonitoredTrainingSession` object.

```
log_vals = {
    'loss': model_loss,

    'step': global_step
}
log_hook = tf.train.LoggingTensorHook(log_vals, every_n_iter=10)
nan_hook = tf.train.NanTensorHook(model_loss)
hooks = [nan_hook, log_hook]
with tf.train.MonitoredTrainingSession(
    hooks=hooks) as sess:
    while not sess.should_stop():
        sess.run(train_op)
```

In the example above, we specified that the training will log the loss and training step every 10 iterations, as well as handle NaN loss. Note that we pass in the logging and NaN hooks as a list with the `hooks` keyword argument.

The `should_stop` function returns a boolean value representing whether the training should stop. It will return true if it reaches the end of the dataset, catches an error (e.g. Nan loss), or a kill signal is sent from the keyboard with `CTRL+C` or `CMD+C`. Using `should_stop` in a while loop will check whether we continue training after each iteration.

D. Checkpoints

One of the most important utilities provided by `MonitoredTrainingSession` is the ability to create a *checkpoint* directory. A checkpoint refers to a saved model state after a specific training iteration. We can save multiple checkpoints and store them in the checkpoint directory, which will also contain the events file used for TensorBoard.

Below we show how to use a `MonitoredTrainingSession` to save the model state as a checkpoint.

```
log_vals = {
    'loss': model_loss,
    'step': global_step
}
log_hook = tf.train.LoggingTensorHook(log_vals, every_n_iter=10)
nan_hook = tf.train.NanTensorHook(model_loss)
hooks = [nan_hook, log_hook]
ckpt_dir = 'my_model'
with tf.train.MonitoredTrainingSession(
    checkpoint_dir=ckpt_dir,
    hooks=hooks) as sess:
    while not sess.should_stop():
        sess.run(train_op)
```

In the example above, we saved the model checkpoints in the *my_model* directory, by setting the `checkpoint_dir` keyword argument accordingly. Notice that when we specify a checkpoint directory, the `global_step/sec` metric is also logged, since it is tracked by default in TensorBoard.

If we run training again using the same checkpoint directory, it will restore the model state from the most recent checkpoint. This is extremely helpful since it allows us to stop and start training at different times, without losing any progress.

E. Model code

In the `ClassificationModel`, we use the `MonitoredTrainingSession` object to run training and save the model state to a checkpoint. The finished `run_model_training` code is shown below:

```
def run_model_training(self, input_data, labels, hidden_layers, batch_size, num_epochs, ckpt_
    self.global_step = tf.train.get_or_create_global_step()
    dataset = self.dataset_from_numpy(input_data, batch_size,
        labels=labels, num_epochs=num_epochs)
    iterator = dataset.make_one_shot_iterator()
    inputs, labels = iterator.get_next()
    self.run_model_setup(inputs, labels, hidden_layers, True)
    self.add_to_tensorboard(inputs)
    log_vals = {'loss': self.loss, 'step': self.global_step}
    logging_hook = tf.train.LoggingTensorHook(
        log_vals, every_n_iter=1000)
    nan_hook = tf.train.NanTensorHook(self.loss)
    hooks = [nan_hook, logging_hook]
    with tf.train.MonitoredTrainingSession(
        checkpoint_dir=ckpt_dir,
        hooks=hooks) as sess:
        while not sess.should_stop():
            sess.run(self.train_op)
```

Checkpoint

Understand how checkpoints are used to save and load TensorFlow models.

Chapter Goals:

- Understand how the checkpoint directory is structured
- Learn how to restore model parameters from a checkpoint

A. Checkpoint directory

After running training with a checkpoint directory, it will contain several files. An example checkpoint directory, named `my_model`, is shown below.

```
ls my_model
```

The `.pbtxt` file represents the entire computation graph stored in human readable text format. The `.tfevents` file is the events file for TensorBoard (note that the longer file suffix, which contains the local machine's ID, is omitted).

The actual saved model state at a particular training step consists of the following three files:

- `.data`: One or more files containing the values for the model's parameters. Larger models may require more `.data` files.
- `.index`: Metadata descriptions for how to find a particular tensor in the `.data` file(s).
- `.meta`: Represents the non-human readable saved graph structure. This file can be used to restore the computation graph.

The `checkpoint` file lists which checkpoint to use when restoring parameters, as well as all the possible checkpoints available.

```
cat my_model/checkpoint
```

Contents of the checkpoint file. The first line specifies the checkpoint to use, while the remaining lines list all the available checkpoints.

B. Saving parameters

The traditional method to save and restore parameters in TensorFlow is to use the `tf.train.Saver` object.

To save the parameters of a given TensorFlow session, use `tf.train.Saver.save`. This function has two required arguments: the current session and the path to which the file will be saved.

It has many keyword arguments, one of which is `global_step`, which determines the number tacked on to the back of the file name.

```
# sess is a tf.Session object
# 'my-model' is the filepath
# global_step
saver.save(sess, 'my-model', global_step=1000)
# checkpoint filename will be 'my-model-1000'
# the file will be in the current working directory
```

C. Restoring parameters

When we use `MonitoredTrainingSession` to resume training, it automatically restores parameters from the `model_checkpoint_path` specified in the `checkpoint` file. However, if we want to evaluate the model or use it for predictions, we need another way to restore the parameters.

The `tf.train.Saver.restore` function is the way to do that.

```
logits = tf.layers.dense(inputs, 1)
saver = tf.train.Saver()
ckpt = tf.train.get_checkpoint_state('my_model')
if ckpt is not None:  # Check if has checkpoint file
  sess = tf.Session()
  saver.restore(sess, ckpt.model_checkpoint_path)
  sess.run(logits)
```

In the example above, we obtain the checkpoint state of `my_model` with the

`tf.train.get_checkpoint_state` function. The function returns a `CheckpointState` object which contains the properties `model_checkpoint_path` and `all_checkpoint_paths`. The former represents the checkpoint to use, while the latter represents the list of all checkpoints. Note that if the checkpoint file is not present in the checkpoint directory, the function returns `None`.

The `Saver` object contains the `restore` function, which restores the checkpoint from the path given in the second argument. The first argument is a `tf.Session` object that we use to execute the restoration.

We can use the `save` function to save the computation graph's parameters. It takes in the same two required arguments as `restore`, and saves the parameters to the directory passed in as the second argument.

Evaluation

Learn how to evaluate a pre-trained model stored in a checkpoint.

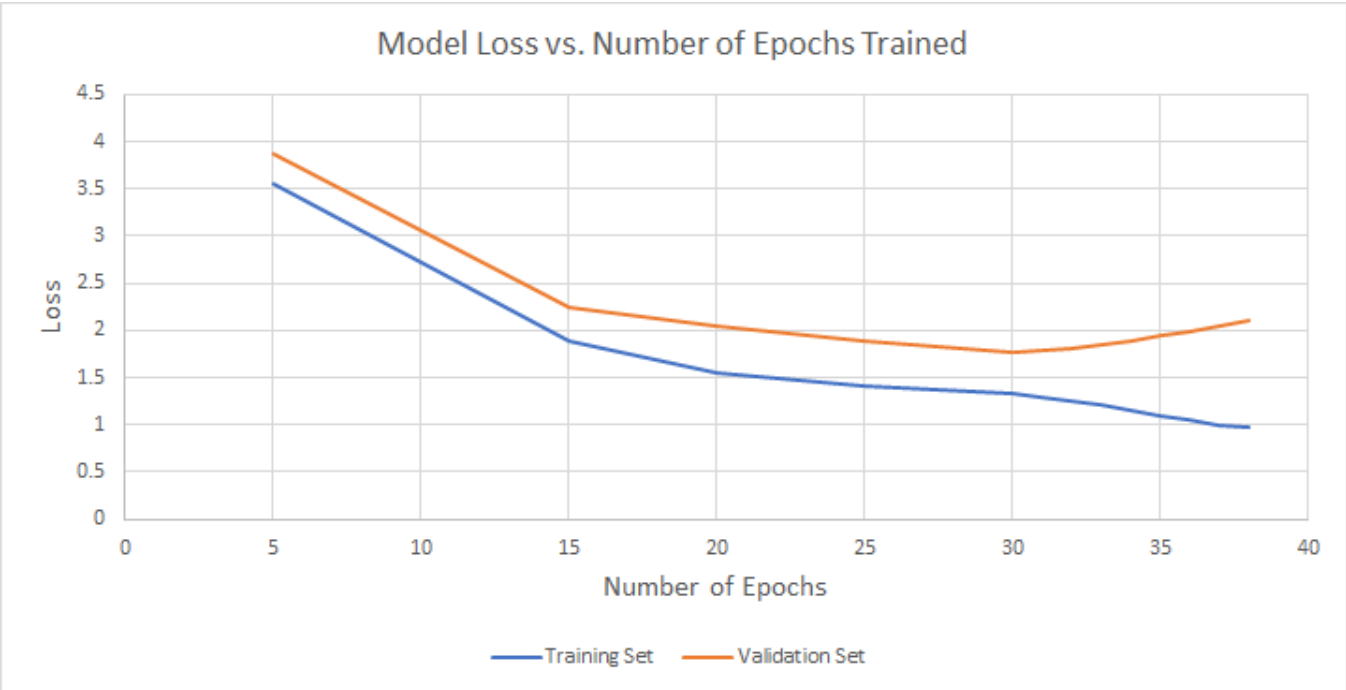
Chapter Goals:

- Understand the difference between training, validation, and test sets
- Learn how to evaluate a trained machine learning model

A. Training vs. evaluation

To measure how well our model has been trained, we evaluate it on datasets other than the training set. The datasets used for evaluation are known as the *validation* and *test* sets. Note that we don't shuffle or repeat the evaluation datasets, since those are techniques used specifically to improve training.

The validation set is used to evaluate a model in between training runs. We use the validation set to tweak certain hyperparameters for a model (such as learning rate or batch size) in order to make sure training continues smoothly. We also use the validation set to detect model overfitting, so we can stop training if overfitting is detected.



Example of overfitting a model on the training set. Overfitting is a common occurrence when training a complex model for too long.

training a complex model for too long.

Overfitting occurs when we train a model (usually a relatively complex model) for too long on the training set, resulting in a decreased ability to generalize well to other datasets. In the above plot, overfitting occurs around the 32nd epoch of training, when the loss on the validation set begins to increase.

The test set is used to evaluate the final version of a model, after it is completely done training. Evaluating on the test set lets us know how well our model performs on its given task.

B. Evaluation metrics

There are a variety of different metrics that can be used for evaluation, depending on the application that a model is built for. However, a universal evaluation metric for machine learning models is the loss. Since every machine learning model is trained to minimize some loss metric, it is natural to use that loss metric during evaluation.

Another commonly used evaluation metric for classification models is accuracy. This refers to the fraction of dataset observations that a machine learning model can label with the correct class. While we train a classification model by minimizing the loss (normally cross entropy), the true goal is to increase model accuracy when classifying new data.

Time to Code!

In this chapter you'll be completing the `evaluate_saved_model` function, which restores a classification model from a checkpoint and then runs evaluation on the model.

First, we get the checkpoint state from the checkpoint directory and make sure the checkpoint file is there.

Set `ckpt` equal to `tf.train.get_checkpoint_state` applied with `ckpt_dir` as the only argument.
Then create an `if` block that checks if `ckpt` is not `None`.

If the checkpoint state is correct, we can set up a `Saver` object and restore the model parameters.

Inside the `if` block, set `saver` equal to a `tf.train.Saver` object, initialized with no arguments.
Then call `saver.restore` with `sess` and `ckpt.model_checkpoint_path` as the two input arguments.

The evaluation metrics we return are the model accuracy and loss. We'll use the input argument `sess` (a `tf.Session` object) to extract the metrics.

Inside the `if` block, set `eval_metrics` equal to `sess.run` with the tuple `(self.accuracy, self.loss)` as the only argument.
Then return `eval_metrics`.

```
import numpy as np
import tensorflow as tf

class ClassificationModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Run model evaluation
    def evaluate_saved_model(self, sess, ckpt_dir):
        # CODE HERE
        pass

    # See the "Efficient Data Processing Techniques" section for details
    def dataset_from_numpy(self, input_data, batch_size, labels=None, is_training=True, num_e
        dataset_input = input_data if labels is None else (input_data, labels)
        dataset = tf.data.Dataset.from_tensor_slices(dataset_input)
        if is_training:
            dataset = dataset.shuffle(len(input_data)).repeat(num_epochs)
        return dataset.batch(batch_size)

    # See the "Machine Learning for Software Engineers" course on Educative
    def run_model_setup(self, inputs, labels, hidden_layers, is_training, calculate_accuracy=
        layer = inputs
        for num_nodes in hidden_layers:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.probs = tf.nn.softmax(logits, name='probs')
        self.predictions = tf.argmax(
            self.probs, axis=-1, name='predictions')
        if calculate_accuracy:
            class_labels = tf.argmax(labels, axis=-1)
            is_correct = tf.equal(
                self.predictions, class_labels)
            is_correct_float = tf.cast(
                is_correct,
                tf.float32)
            self.accuracy = tf.reduce_mean(
                is_correct_float)
```

```
        is_correct_float)
    if labels is not None:
        labels_float = tf.cast(
            labels, tf.float32)
        cross_entropy = tf.nn.softmax_cross_entropy_with_logits_v2(
            labels=labels_float,
            logits=logits)
        self.loss = tf.reduce_mean(
            cross_entropy)
    if is_training:
        adam = tf.train.AdamOptimizer()
        self.train_op = adam.minimize(
            self.loss, global_step=self.global_step)

# Run training of the classification model
def run_model_training(self, input_data, labels, hidden_layers, batch_size, num_epochs, c
    self.global_step = tf.train.get_or_create_global_step()
    dataset = self.dataset_from_numpy(input_data, batch_size,
        labels=labels, num_epochs=num_epochs)
    iterator = dataset.make_one_shot_iterator()
    inputs, labels = iterator.get_next()
    self.run_model_setup(inputs, labels, hidden_layers, True)
    tf.summary.scalar('accuracy', self.accuracy)
    tf.summary.histogram('inputs', inputs)
    log_vals = {'loss': self.loss, 'step': self.global_step}
    logging_hook = tf.train.LoggingTensorHook(
        log_vals, every_n_iter=1000)
    nan_hook = tf.train.NanTensorHook(self.loss)
    hooks = [nan_hook, logging_hook]
    with tf.train.MonitoredTrainingSession(
        checkpoint_dir=ckpt_dir,
        hooks=hooks) as sess:
        while not sess.should_stop():
            sess.run(self.train_op)
```

Save For Inference

Save a condensed model to be used for inference on real-time data.

Chapter Goals:

- Learn how to save a model for inference

A. Deploying a model

As mentioned in chapter 4, a saved model checkpoint consists of three files: `.data`, `.index`, and `.meta`. Since the `.meta` file contains the entire computation graph structure, which includes all the data in the training dataset, it can get quite large. The large file size becomes an issue when deploying an *inference* model.

An inference model is a fully trained and evaluated model used to make predictions on real-time data. When we deploy an inference model for production, we don't usually deploy the code used to build the model, either for proprietary reasons or because there are too many auxiliary code files. When we don't have the code that sets up the inference graph, we need a separate file that specifies the computation graph's structure.

B. Inference graph

The main issue with the `.meta` file is that it contains many unnecessary portions of the computation graph, with respect to inference. For inference, we only need a `tf.placeholder` to represent the input data. We also don't need any parts of the computation graph specific to training, such as the loss calculation or dataset.

So instead of using a training checkpoint for the inference model, we create a bare-bones computation graph, consisting only of the input placeholder and the computations necessary to obtain a prediction.

If the model just finished training (meaning the training computation graph is still in memory), it's necessary to use `tf.reset_default_graph` prior to building

the inference graph, in order to avoid graph conflicts.

```
import tensorflow as tf

inputs = tf.placeholder(
    tf.float32, shape=(None, 3), name='inputs')
logits = tf.layers.dense(inputs, 1, name='logits')
try:
    logits = tf.layers.dense(inputs, 1, name='logits')
except ValueError: # Need to reset graph
    tf.reset_default_graph()
    inputs = tf.placeholder(
        tf.float32, shape=(None, 3), name='inputs')
    logits = tf.layers.dense(inputs, 1, name='logits')
    print(logits)
```

Using tf.reset_default_graph to clear the previous computation graph from memory.

C. Saving the model

We save the inference model using `tf.saved_model.simple_save`. The function's first argument is a `tf.Session` object and the second argument is the path to the directory where we save the inference model. Note that the directory with which we save the inference model must not already exist.

The third argument is a dictionary containing the input tensor(s) as values, with string labels as keys. The fourth required argument is also a dictionary with string keys, but for the output tensor(s), e.g. the model prediction.

The function will save a file called `saved_model.pb` and a directory called `variables` in the specified directory.

```
ls inference_dir
```

In the example above, `inference_dir` is the directory where `tf.saved_model.simple_save` saved the inference model. The `saved_model.pb` file contains the bare-bones computation graph, and is much smaller than the corresponding `.meta` file. The `variables` directory contains the model's saved parameters.

In the next chapter, you'll learn how to restore the inference model and make predictions.

Time to Code!

In this chapter you'll complete the `save_inference_graph` function, which saves the model's computation graph for inference. The function is already filled with code that restores the model state from a checkpoint.

The input dictionary for the inference graph contains `input_placeholder` as its only value, which represents the input data for the inference graph.

Set `input_dict` equal to a dictionary with a single key, `'inputs'`, that maps to `input_placeholder`.

The output dictionary for the inference graph contains `self.predictions` as its only value. The corresponding key is `'predictions'`.

Set `output_dict` equal to a dictionary consisting of the specified key-value pair.

After creating the dictionaries for the inference graph's input and output, we can save the model using `tf.saved_model.simple_save`.

Call the specified function with `sess`, `export_dir`, `input_dict`, and `output_dict` as the four input arguments.

```
import numpy as np
import tensorflow as tf

class ClassificationModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Save the model's computation graph for inference
    def save_inference_graph(self, sess, ckpt_dir, input_placeholder, export_dir):
        ckpt = tf.train.get_checkpoint_state(ckpt_dir)
        if ckpt is not None:
            saver = tf.train.Saver()
            saver.restore(sess, ckpt.model_checkpoint_path)
            # CODE HERE
            pass

    # See the "Efficient Data Processing Techniques" section for details
    def dataset_from_numpy(self, input_data, batch_size, labels=None, is_training=True, num_e
```

```
        dataset_input = input_data if labels is None else (input_data, labels)
        dataset = tf.data.Dataset.from_tensor_slices(dataset_input)
        if is_training:

            dataset = dataset.shuffle(len(input_data)).repeat(num_epochs)
        return dataset.batch(batch_size)

    # See the "Machine Learning for Software Engineers" course on Educative
    def run_model_setup(self, inputs, labels, hidden_layers, is_training, calculate_accuracy=
        layer = inputs
        for num_nodes in hidden_layers:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.probs = tf.nn.softmax(logits, name='probs')
        self.predictions = tf.argmax(
            self.probs, axis=-1, name='predictions')
        if calculate_accuracy:
            class_labels = tf.argmax(labels, axis=-1)
            is_correct = tf.equal(
                self.predictions, class_labels)
            is_correct_float = tf.cast(
                is_correct,
                tf.float32)
            self.accuracy = tf.reduce_mean(
                is_correct_float)
        if labels is not None:
            labels_float = tf.cast(
                labels, tf.float32)
            cross_entropy = tf.nn.softmax_cross_entropy_with_logits_v2(
                labels=labels_float,
                logits=logits)
            self.loss = tf.reduce_mean(
                cross_entropy)
        if is_training:
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)

    # Run training of the classification model
    def run_model_training(self, input_data, labels, hidden_layers, batch_size, num_epochs, c
        self.global_step = tf.train.get_or_create_global_step()
        dataset = self.dataset_from_numpy(input_data, batch_size,
            labels=labels, num_epochs=num_epochs)
        iterator = dataset.make_one_shot_iterator()
        inputs, labels = iterator.get_next()
        self.run_model_setup(inputs, labels, hidden_layers, True)
        tf.summary.scalar('accuracy', self.accuracy)
        tf.summary.histogram('inputs', inputs)
        log_vals = {'loss': self.loss, 'step': self.global_step}
        logging_hook = tf.train.LoggingTensorHook(
            log_vals, every_n_iter=1000)
        nan_hook = tf.train.NanTensorHook(self.loss)
        hooks = [nan_hook, logging_hook]
        with tf.train.MonitoredTrainingSession(
            checkpoint_dir=ckpt_dir,
            hooks=hooks) as sess:
            while not sess.should_stop():
                sess.run(self.train_op)
```

Predictions

Restore an inference model and make predictions on an input dataset.

Chapter Goals:

- Learn how to restore an inference model and retrieve specific tensors from the computation graph
- Implement a function that makes predictions using a saved inference model

A. Restoring the model

To restore an inference model, we use the `tf.saved_model.loader.load` function. This function restores both the inference graph as well as the inference model's parameters.

Since the function's first argument is a `tf.Session` object, it's a good idea to restore the inference model within the scope of a particular `tf.Session`.

```
import tensorflow as tf

tags = [tf.saved_model.tag_constants.SERVING]
model_dir = 'inference_model'
with tf.Session(graph=tf.Graph()) as sess:
    tf.saved_model.loader.load(sess, tags, model_dir)
```

Restoring an inference model inside the scope of the tf.Session object, sess.

The second argument for `tf.saved_model.loader.load` is a list of tag constants. For inference, we use the `SERVING` tag. The function's third argument is the path to the saved inference model's directory.

In the example above, we restored the inference model within the scope of a `tf.Session` object named `sess`. Note that `sess` is initialized with a new, empty computation graph in `tf.Graph()`. This ensures that there are no graph conflicts, similar to `tf.reset_default_graph`, but also that it doesn't erase the graph outside the scope of `sess`.

B. Retrieving tensors

After restoring the inference model, the inference graph is represented by `sess.graph`, which is a `Graph` object. We use the `Graph` object's `get_tensor_by_name` function to retrieve specific tensors from the computation graph.

```
import tensorflow as tf

tags = [tf.saved_model.tag_constants.SERVING]
model_dir = 'inference_model'
with tf.Session(graph=tf.Graph()) as sess:
    tf.saved_model.loader.load(sess, tags, model_dir)
    preds = sess.graph.get_tensor_by_name('predictions:0')
```

Retrieving the predictions tensor, preds, with get_tensor_by_name.

The input argument for `get_tensor_by_name` is the name of the tensor we wish to retrieve. In the above example, the tensor name has `:0` as a suffix. Whenever we retrieve a single tensor from the computation graph, we need to add the `:0` suffix.

Time to Code!

In this chapter you'll be completing the `make_predictions` function, which makes predictions using a saved inference graph. Specifically, you'll be creating the helper function `load_inference_parts`, which loads the inference graph and gets the necessary tensors.

We can load the saved inference model using the function `tf.saved_model.loader.load`. The tag list we use to load the inference model is a singleton list containing `tf.saved_model.tag_constants.SERVING`.

Call the loading function with `sess`, the specified tag list, and `saved_model_dir` as the three input arguments.

Next, we'll get the input placeholder and the predictions tensor from the loaded computation graph. The computation graph corresponds to `sess.graph`, and we use `sess.graph.get_tensor_by_name` to retrieve a specific tensor based on its name.

Set `inputs` equal to the tensor retrieving function with

`'inference_input:0'` **as the only argument.**

Set `predictions_tensor` equal to the tensor retrieving function with `'predictions:0'` as the only argument.

Since the model probabilities are used to calculate the predictions, the probabilities tensor is part of the computation graph. Therefore, we can retrieve the tensor using its name.

Set `probs_tensor` equal to the tensor retrieving function with `'probs:0'` as the only argument.

Return a tuple containing `inputs` as the first element, `predictions_tensor` as the second element, and `probs_tensor` as the third element.

```
import numpy as np
import tensorflow as tf

class ClassificationModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Helper for make_predictions
    def load_inference_parts(self, sess, saved_model_dir):
        # CODE HERE
        pass

    # Make predictions with the inference model
    def make_predictions(self, saved_model_dir, input_data):
        with tf.Session(graph=tf.Graph()) as sess:
            inputs, predictions_tensor, probs_tensor = self.load_inference_parts(sess, saved_
            predictions, probs = sess.run((predictions_tensor, probs_tensor), feed_dict={inpu
        return predictions, probs

    # See the "Efficient Data Processing Techniques" section for details
    def dataset_from_numpy(self, input_data, batch_size, labels=None, is_training=True, num_e
        dataset_input = input_data if labels is None else (input_data, labels)
        dataset = tf.data.Dataset.from_tensor_slices(dataset_input)
        if is_training:
            dataset = dataset.shuffle(len(input_data)).repeat(num_epochs)
        return dataset.batch(batch_size)

    # See the "Machine Learning for Software Engineers" course on Educative
    def run_model_setup(self, inputs, labels, hidden_layers, is_training, calculate_accuracy=
        layer = inputs
        for num_nodes in hidden_layers:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.probs = tf.nn.softmax(logits, name='probs')
        self.predictions = tf.argmax(
            self.probs, axis=-1, name='predictions')
        if calculate_accuracy:
```

```
        if calculate_accuracy:
            class_labels = tf.argmax(labels, axis=-1)
            is_correct = tf.equal(
                self.predictions, class_labels)
            is_correct_float = tf.cast(
                is_correct,
                tf.float32)
            self.accuracy = tf.reduce_mean(
                is_correct_float)
        if labels is not None:
            labels_float = tf.cast(
                labels, tf.float32)
            cross_entropy = tf.nn.softmax_cross_entropy_with_logits_v2(
                labels=labels_float,
                logits=logits)
            self.loss = tf.reduce_mean(
                cross_entropy)
        if is_training:
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)

    # Run training of the classification model
    def run_model_training(self, input_data, labels, hidden_layers, batch_size, num_epochs, c
        self.global_step = tf.train.get_or_create_global_step()
        dataset = self.dataset_from_numpy(input_data, batch_size,
            labels=labels, num_epochs=num_epochs)
        iterator = dataset.make_one_shot_iterator()
        inputs, labels = iterator.get_next()
        self.run_model_setup(inputs, labels, hidden_layers, True)
        tf.summary.scalar('accuracy', self.accuracy)
        tf.summary.histogram('inputs', inputs)
        log_vals = {'loss': self.loss, 'step': self.global_step}
        logging_hook = tf.train.LoggingTensorHook(
            log_vals, every_n_iter=1000)
        nan_hook = tf.train.NanTensorHook(self.loss)
        hooks = [nan_hook, logging_hook]
        with tf.train.MonitoredTrainingSession(
            checkpoint_dir=ckpt_dir,
            hooks=hooks) as sess:
            while not sess.should_stop():
                sess.run(self.train_op)
```

Below is an example plot of some predictions (run the show_plot() function). The points in this plot represent the winning percentages of MLB teams in 2017.

The plot's *z*-axis corresponds to winning percentage, the *y*-axis corresponds to runs per game, and the *x*-axis corresponds to batting average.

The inference model was trained on runs per game and batting average to classify a team into one of three classes:

- *Poor*: Winning percentage below 0.450 (73 wins)
- *Average*: Winning percentage between 0.450 and 0.543 (88 wins)
- *Good*: Winning percentage above 0.543

The winning percentage thresholds are represented by the two planes in the 3-D plot. Correct predictions by the model are marked in blue, while incorrect predictions are marked in red.

If you hover over any of the points, it will give you information on the model's prediction and probability per class.

```
show_plot()
```

Regression

Understand the difference between regression and classification models.

Chapter Goals:

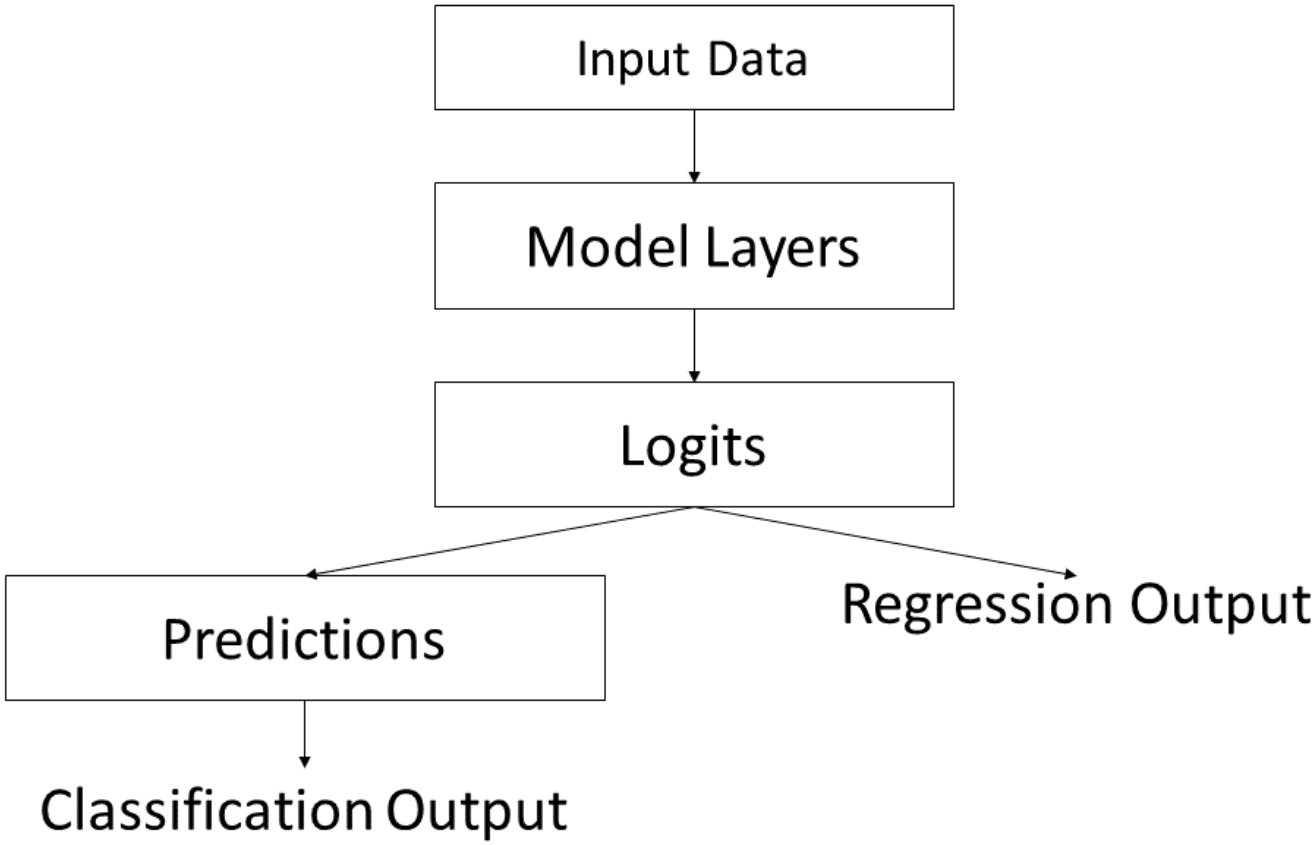
- Learn how to use a neural network for regression
- Set up the training architecture for a regression model

A. Classification vs. regression

The model we've been using so far is built for classification. This means it classifies input data into one of many different classes.

While classification is one of the most important tasks that can be done by a neural network, another common application of neural networks is *regression*. Rather than assigning the input data to a particular class, a regression model will instead output a real number value based on the input data. Regression tasks can range from predicting stock prices to predicting the future height of a child.

Classification models obtain probabilities and class predictions from the calculated logits. A regression model can just return the logits, since the logits represent the real-valued output of the model's layers.



Classification output vs. regression output.

For the remainder of this course you'll be creating the regression model that can be used for sales predictions.

B. L2 loss

When training a classification model, the loss function we use is the cross entropy loss. For regression models, we normally use the L2 loss, which is a loss function derived from the L2-norm.

In TensorFlow, the L2 loss is calculated by the `tf.nn.l2_loss` function. Note that the TensorFlow implementation omits the square root used in the L2-norm and also divides the value by 2. This makes it easier to calculate the gradient for weight optimization.

We normally use L2 loss on the subtracted difference between two scalars or tensors. The L2 loss represents how "close" the two values are.

```
import tensorflow as tf
t1 = tf.constant([9.3, 95.4, -0.3])
t2 = tf.constant([9.35, 95.4, -0.28])
t3 = tf.constant([-80.4, -95.1, 78])
with tf.Session() as sess:
```

```
with tf.Session() as sess:
  print(sess.run(tf.nn.l2_loss(t1 - t2)))
  print(sess.run(tf.nn.l2_loss(t1 - t3)))
```

An L2 loss near 0 means the two values are very close, while a large L2 loss means the values are not close at all. In our example, the elements of `t1` and `t2` are extremely close, so their L2 loss is tiny. On the other hand, `t1` and `t3` are not close at all, so their L2 loss is large.

For regression models, we calculate the L2 loss between the model's output (i.e. logits) and the labels for the input data.

Time to Code!

In this chapter you'll be completing part of the `regressor_fn` function, which is the main function for the regression model (`RegressionModel`) that will be created in this section of the course.

Specifically, you'll be completing the helper function `set_predictions_and_loss`, which sets the regression predictions and loss (if necessary).

The `logits` input argument, which represents the regression model's logits, has shape `(batch_size, 1)`. However, we want our model predictions to have the same shape as `labels`, which is `(batch_size,)`.

In this case, we use `tf.squeeze` to remove the size 1 second dimension from `logits`. The new 1-D tensor corresponds to our model predictions.

Set `self.predictions` equal to `tf.squeeze` with `logits` as the only argument.

If `labels` is specified, we want to compute the regression loss. The loss function used for regression is `tf.nn.l2_loss`.

If `labels` is not `None`, set `self.loss` equal to `tf.nn.l2_loss` applied to the difference between `labels` and `self.predictions`.

```
import numpy as np
import tensorflow as tf
```

```
class RegressionModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Helper for regressor_fn
    def set_predictions_and_loss(self, logits, labels):
        # CODE HERE
        pass

    # The function for the regression model
    def regressor_fn(self, features, labels, mode, params):
        inputs = tf.feature_column.input_layer(features, params['feature_columns'])
        layer = inputs
        for num_nodes in params['hidden_layers']:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.set_predictions_and_loss(logits, labels)
        if mode == tf.estimator.ModeKeys.TRAIN:
            self.global_step = tf.train.get_or_create_global_step()
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)
        if mode == tf.estimator.ModeKeys.EVAL:
            pass
        if mode == tf.estimator.ModeKeys.PREDICT:
            pass
```

EstimatorSpec

Use the EstimatorSpec object to organize training, evaluation, and prediction.

Chapter Goals:

- Learn how to evaluate a regression model
- Use the `EstimatorSpec` object to organize results from training, evaluation, and prediction

A. Regression evaluation

Unlike classification models, we can't use the accuracy metric to evaluate regression. Since the output of regression models is a real number, rather than a class prediction, there's no definite way to say what is "correct" or "incorrect".

However, we can tell how good a model output is based on its distance from the corresponding label. For example, if the label for some data observation was 0.2 and the model returned 0.199, the prediction is excellent. On the other hand, if the model returned 812.11, the prediction is likely very poor.

The metric that corresponds to this idea is mean squared error (MSE). The MSE is very similar to the L2-norm, in that both are based on the squared difference between labels and predictions.

In TensorFlow, we obtain the MSE metric using `tf.metrics.mean_squared_error`. The function takes in the labels and model predictions as its two required arguments.

```
import tensorflow as tf

mse_metric = tf.metrics.mean_squared_error(labels, predictions)
assert isinstance(mse_metric, tuple) and len(mse_metric) == 2
```

Obtaining the MSE metric using tf.metrics.mean_squared_error. Note that the function's output is a tuple containing a tensor and an update operation.

The first element of the output tuple is a tensor representing the overall MSE after each evaluation step. The second tuple element is an operation that's used to update the overall MSE after each evaluation step. This gives us a cumulative MSE when we finish evaluating all the data observations.

B. Using `EstimatorSpec`

The upcoming chapters deal with TensorFlow's `Estimator` API, which encapsulates training, evaluating, and predicting into one compact object. However, in order to use the `Estimator` object, we need to first organize the model results in an `EstimatorSpec`.

The `EstimatorSpec` object is initialized with a single required argument, called `mode`. The `mode` can take one of three values:

- `tf.estimator.ModeKeys.TRAIN`
- `tf.estimator.ModeKeys.EVAL`
- `tf.estimator.ModeKeys.PREDICT`

The keyword arguments required to initialize the `EstimatorSpec` will differ depending on the `mode`.

Time to Code!

In this chapter you'll be completing the `regressor_fn` by creating helper functions for the `EVAL` and `PREDICT` blocks (the code for the `TRAIN` block is already filled in.

The first helper function you'll create is the `eval_regressor` function, for the `EVAL` block. It applies mean squared error as the evaluation metric.

Then set `mse_metric` equal to `tf.metrics.mean_squared_error` with `labels` as the first argument and `self.predictions` as the second argument.

Set `eval_metric` equal to a dictionary with `'mse'` as the only key and `mse_metric` as the corresponding value.

When initializing `EstimatorSpec` in `EVAL` mode, the required keyword argument is `loss`, meaning that the evaluation will always use the model loss as a metric.

as a metric.

We can add more evaluation metrics through the `eval_metric_ops` keyword argument. The argument takes in a dictionary, which maps string names for each metric to tuple values. Each tuple contains a tensor and update operation (e.g. `mse_metric`).

Set `estimator_spec` equal to `tf.estimator.EstimatorSpec` initialized with `mode` as the required argument. Use `self.loss` for the `loss` keyword argument and `eval_metric` for the `eval_metric_ops` keyword argument.

Then return `estimator_spec`.

```
import numpy as np
import tensorflow as tf

class RegressionModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Helper for regressor_fn
    def eval_regressor(self, mode, labels):
        # CODE HERE
        pass

    # Helper from previous chapter
    def set_predictions_and_loss(self, logits, labels):
        self.predictions = tf.squeeze(logits)
        if labels is not None:
            self.loss = tf.nn.l2_loss(labels - self.predictions)

    # The function for the regression model
    def regressor_fn(self, features, labels, mode, params):
        inputs = tf.feature_column.input_layer(features, params['feature_columns'])
        layer = inputs
        for num_nodes in params['hidden_layers']:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.set_predictions_and_loss(logits, labels)
        if mode == tf.estimator.ModeKeys.TRAIN:
            self.global_step = tf.train.get_or_create_global_step()
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)
            return tf.estimator.EstimatorSpec(mode,
                loss=self.loss, train_op=self.train_op)
        if mode == tf.estimator.ModeKeys.EVAL:
            return self.eval_regressor(mode, labels)
        if mode == tf.estimator.ModeKeys.PREDICT:
            pass
```

The second helper function you'll create is the `predict_regressor` function, for the `PREDICT` block. It sets up the regression predictions for the model.

The `features` argument contains the names for each data observation. These names will be returned along with the model predictions, so we can easily identify each data observation.

Create a dictionary with two keys, `'predictions'` and `'names'`, which map to `self.predictions` and `features['name']`, respectively. Store the dictionary in a variable named `prediction_info`.

When initializing `EstimatorSpec` in `PREDICT` mode, the required keyword argument is `predictions`. This represents a dictionary that contains the output values for prediction mode. The keys are string names used to identify each output, while the values are the output tensors.

Set `estimator_spec` equal to `tf.estimator.EstimatorSpec` with `mode` as the required argument and `prediction_info` as the `predictions` keyword argument.

Then return `estimator_spec`.

```
import numpy as np
import tensorflow as tf

class RegressionModel(object):
    def __init__(self, output_size):
        self.output_size = output_size

    # Helper for regressor_fn
    def predict_regressor(self, mode, features):
        # CODE HERE
        pass

    # Helper from previous chapter
    def set_predictions_and_loss(self, logits, labels):
        self.predictions = tf.squeeze(logits)
        if labels is not None:
            self.loss = tf.nn.l2_loss(labels - self.predictions)

    # The function for the regression model
    def regressor_fn(self, features, labels, mode, params):
        inputs = tf.feature_column.input_layer(features, params['feature_columns'])
        layer = inputs
        for num_nodes in params['hidden_layers']:
            layer = tf.layers.dense(layer, num_nodes,
                activation=tf.nn.relu)
```

```
        logits = tf.layers.dense(layer, self.output_size,
            name='logits')
        self.set_predictions_and_loss(logits, labels)

        if mode == tf.estimator.ModeKeys.TRAIN:
            self.global_step = tf.train.get_or_create_global_step()
            adam = tf.train.AdamOptimizer()
            self.train_op = adam.minimize(
                self.loss, global_step=self.global_step)
            return tf.estimator.EstimatorSpec(mode,
                loss=self.loss, train_op=self.train_op)
        if mode == tf.estimator.ModeKeys.EVAL:
            # SEE PREVIOUS EXERCISE
            pass
        if mode == tf.estimator.ModeKeys.PREDICT:
            return self.predict_regressor(mode, features)
```

Estimator

Learn how to use TensorFlow's Estimator API for model development.

Chapter Goals:

- Create an `Estimator` object for the regression model

A. Creating an `Estimator`

TensorFlow's `Estimator` object provides an organized and simple API for model execution. It handles model training, saving and restoring checkpoints, evaluating a model, and making predictions.

To initialize an `Estimator` object, we pass in the model function as a required argument. The model function should follow the same template as `regressor_fn`. The function must return an `ExampleSpec` object, which specifies the model results for training, evaluation, or prediction.

The two main keyword arguments to know are `model_dir` and `params`. The `model_dir` argument represents the directory we save model checkpoints to. The `params` argument represents the values we wish to pass into the model function. The argument should be set to a dictionary, which then corresponds to the model function's `params` argument.

```
import tensorflow as tf
params = {
    'feature_columns': feature_columns,
    'hidden_layers': hidden_layers
}
regressor = tf.estimator.Estimator(
    regressor_fn,
    model_dir=ckpt_dir,
    params=params)
```

Creating the Estimator object for the regression model.

In our example, we initialized the `Estimator` object with `regressor_fn` as its model function. We set the checkpoint directory to `ckpt_dir`, and passed in

the feature columns and number of hidden layers through `params`.

Estimator Train

Use the Estimator API to train a regression model.

Chapter Goals:

- Use an `Estimator` object for training a regression model

A. Training

The `Estimator` object provides a function called `train`, which takes care of all the necessary model training tasks. This includes running the training operation, saving and restoring checkpoints, and initializing variables.

The `train` function takes an input data function as the required argument. The input data function must take in no arguments and return a dataset for the input data and labels. In order to create a function that returns the input dataset without taking in any arguments, we usually use a lambda wrapper around our main dataset function.

```
def make_dataset(filenames, example_spec, batch_size):
    # function code omitted

input_fn = lambda:make_dataset(
    ['data.tfrecords'],
    example_spec,
    16)
```

Creating a lambda wrapper around make_dataset, which returns the dataset containing input data and labels.

We can use specific hooks (e.g. for logging values) by setting the `hooks` keyword argument of `train`. However, the `Estimator` object automatically logs the loss and global step during training, so there is no need to manually log those values with `hooks`. The `Estimator` also detects NaN loss and subsequently stops training if loss becomes NaN.

If we want to specify the number of steps to train on a particular training run, we use the `steps` keyword argument. To instead specify the *maximum*

number of steps to train a model, taking into account how many steps the model has already been trained for, we set the `max_steps` keyword argument.

If `steps` and `max_steps` are both `None` (the default values), the training will continue until it reaches the end of the dataset or we manually stop training with `CTRL+C` or `CMD+C`.

Full code for training the regression model is shown below:

```
def dataset_from_examples(self, filenames, example_spec, batch_size,
    buffer_size=None, use_labels=True, num_epochs=None):
    dataset = tf.data.TFRecordDataset(filenames)
    def _parse_fn(example_bytes):
        parsed_features = tf.parse_single_example(example_bytes, example_spec)
        label = parsed_features['label']
        output_features = [k for k in parsed_features.keys() if k != 'label']
        if use_labels:
            return {k: parsed_features[k] for k in output_features}, label
        return {k: parsed_features[k] for k in output_features}
    dataset = dataset.map(_parse_fn)
    if buffer_size is not None:
        dataset = dataset.shuffle(buffer_size)
    return dataset.repeat(num_epochs).batch(batch_size)

def run_regressor_training(self, ckpt_dir, hidden_layers, feature_columns, filenames,
    example_spec, batch_size, num_examples, num_training_steps=None):
    params = {
        'feature_columns': feature_columns,
        'hidden_layers': hidden_layers
    }
    regressor = tf.estimator.Estimator(
        self.regressor_fn,
        model_dir=ckpt_dir,
        params=params)
    input_fn = lambda:self.dataset_from_examples(filenames, example_spec, batch_size,
        buffer_size=num_examples)
    regressor.train(
        input_fn,
        steps=num_training_steps)
```

Here, `self.regressor_fn` is the function you completed previously for the `RegressionModel` object.

The `dataset_from_examples` function creates the training dataset from serialized Example objects representing rows of the CSV.

Estimator Eval

Use the Estimator API to evaluate the regression model.

Chapter Goals:

- Use an `Estimator` object to evaluate a regression model

A. Evaluation

The `Estimator` object provides a function called `evaluate`, which is used for model evaluation.

Like the `train` function, `evaluate` also takes an input data function as its required argument. The input data function must follow the same format as the one used with `train`, although it can be configured differently (i.e. no shuffling, smaller batch size, etc.). We can also specify the number of steps to run evaluation for using the `steps` keyword argument.

The `evaluate` function will return a dictionary containing the final values for each evaluation metric. Each key is the name of a metric and each value is that metric's value. The dictionary metrics consist of the model loss (corresponding to key `'loss'`), as well as each metric we specified in `eval_metric_ops` when initializing the evaluation `ExampleSpec`. The dictionary will also include the final training iteration of the model, represented by `'global_step'`.

```
eval_dict = regressor.evaluate(
    input_fn, # lambda function
    steps=2)
```

Note that `evaluate` also creates a new `eval` directory inside the model checkpoint directory. In the code above, the `eval` directory is saved as a subdirectory within `model_dir`. The `eval` directory contains a single events file depicting the evaluation run, which can be used with TensorBoard.

Estimator Predict

Use the Estimator API to make predictions with the regression model.

Chapter Goals:

- Learn how to use an `Estimator` object to make regression predictions

A. Prediction

The `Estimator` object provides a function called `predict`, which is used for making model predictions.

Like the `train` and `evaluate` functions, `predict` also takes an input data function as its required argument. However, the input data function for `predict` does not need to return any labels, since we don't return the loss when making model predictions. In fact, it is often the case that there are no actual labels for the predicted data observations.

The `predict` function will return a generator of dictionaries, where each dictionary corresponds to a predicted data observation. The dictionary consists of the output values specified in the model function's prediction `ExampleSpec`.

```
preds = regressor.predict(input_fn)
```

By default, each dictionary contains all the values from the `predictions` argument of the `ExampleSpec`. We can choose to return specific values by setting the `predict_keys` keyword argument of `predict`. The argument takes in a list of strings, corresponding to the names of the values we want returned in each dictionary.

```
preds = regressor.predict(
    input_fn,
    predict_keys=['prediction'])
```

Below is a plot of regression predictions made with `Estimator` (run the show_plot() function). The points represent the 2015 admission rates of 50 randomly chosen colleges.

The plot's *z*-axis corresponds to admission rate, the *y*-axis corresponds to average SAT scores from 2013-2015, and the *x*-axis corresponds to average tuition from 2013-2016.

The regression model was trained on a variety of factors, of which SAT scores and college tuition were the most informative. Its task was to predict a school's 2015 admission rate.

Points representing actual admission rates are marked in blue, while model predictions are marked in green.

If you hover over any of the points, it will tell you the name of the school and either the actual or predicted admission rate. You can also click the grey icon bar at the top right of the plot to adjust camera settings.

```
show_plot()
```

Quiz

1

Which of the following values does TensorBoard automatically track?

COMPLETED 0%

1 of 4